



N mindsTM
Academy





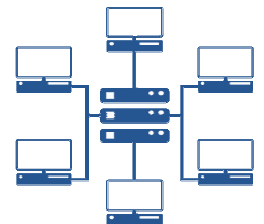
Networking-Workshop

Present By,

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Founder & Senior Technical Trainer

Nminds Academy-Erode



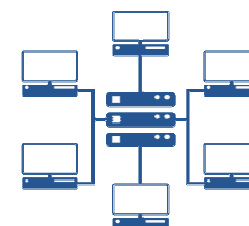
What is Network?

- ✓ Network is an **interconnection of network enabled devices.**

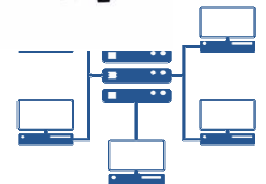
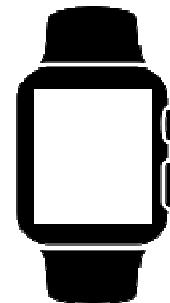
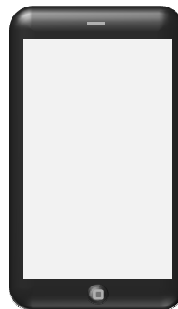
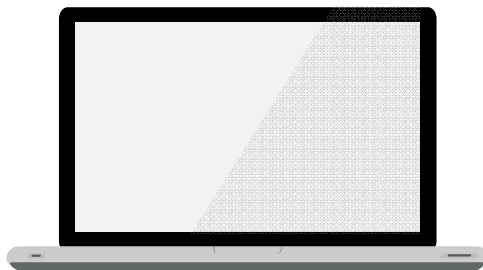
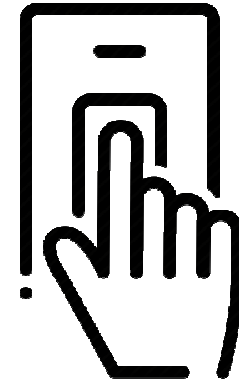
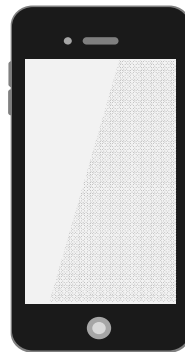
OR

- ✓ Network is a collection of devices for **sharing of resources** and for communication purpose.
- ✓ Here, Sharing of Resources means to share **Sharing of hardware and software resource** that are available simultaneously to **everyone connecting together.**

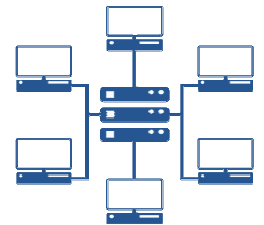
Example: Documents, Printer, Internet etc.



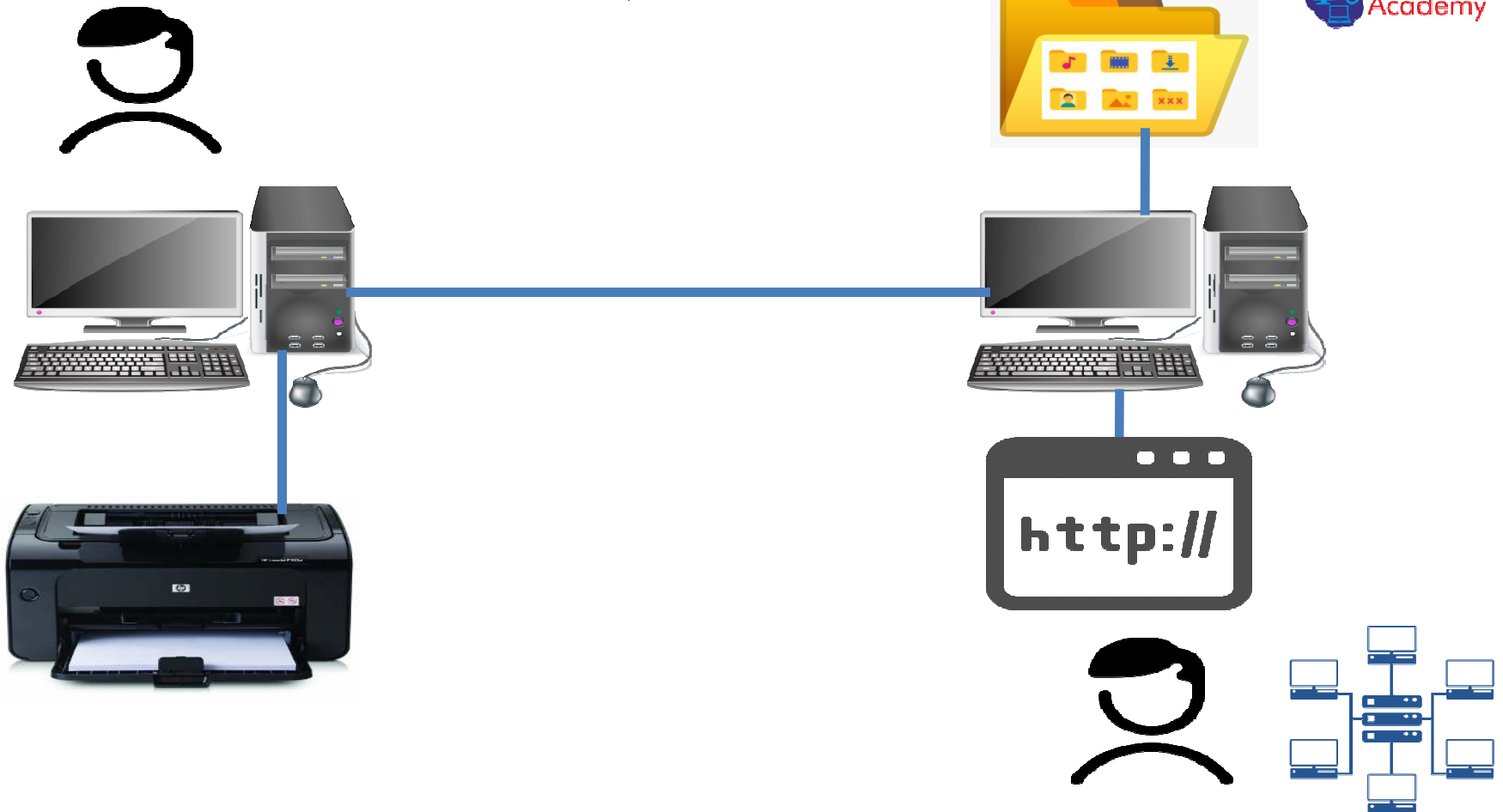
Network Enabled Devices



What is the need of Networking?



How N/W done?



Advantages and Disadvantages of N/W?

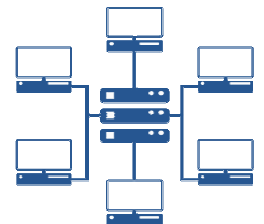


Advantages of Network

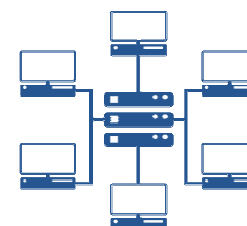
- Communication is faster and easier.
- Cost effective.
- No size limitation.
- Sharing of hardware and software resource.

Disadvantage of Network

- Security should be integrated over the network.
- Need for an Administrator to take care of the network.



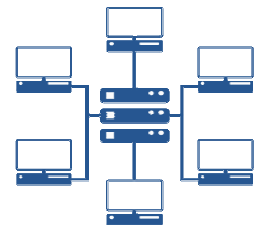
Components required for establishing network



Components required for establishing network?



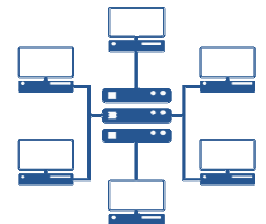
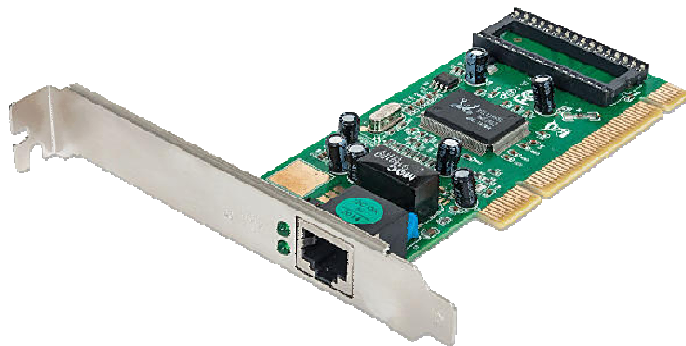
- ✓ Network Enabled Device-Computer/Laptop
- ✓ Media (Wired or Wireless)
- ✓ Networking devices
- ✓ Protocol
- ✓ IP Address



Network Enabled Device-NIC

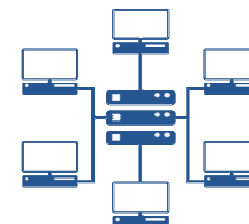


- NIC (Network Interface Card) is a port or interface to connect network from one device with another device.
- All Network enabled device has NIC (Wired) OR Wireless(Wi-Fi Receiver)



NIC-MAC Address

- It has MAC(Medium Access Control) address, this address is used for communication.
- It is an Unique address, cannot change this address.
- It is also known as physical address.
- This address is used to for identification of each network enabled device.



MAC Address?

➤ It has **48 bit hexadecimal address**.

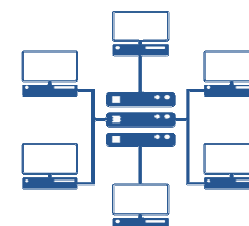
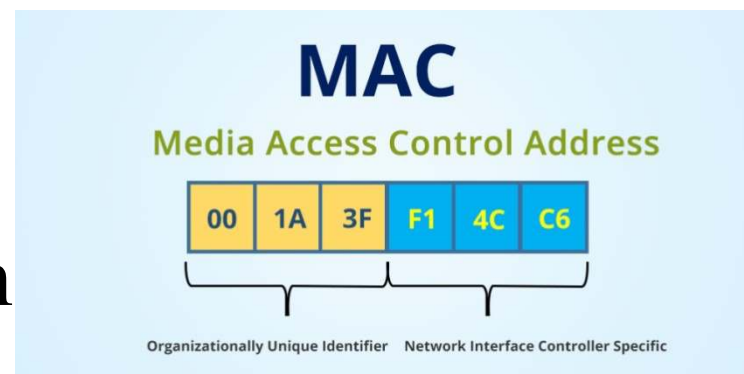
➤ Cmd-getmac

18-60-24-48-B1-37

➤ It is divided into two portion

First 24 bit is called Vendor-Id

Last 24 bit is called Adapter-ID



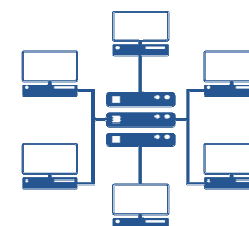
MEDIA

Media

- It establishes communication path to communicate over the network. There are 2 types of media, they are

Wired Media

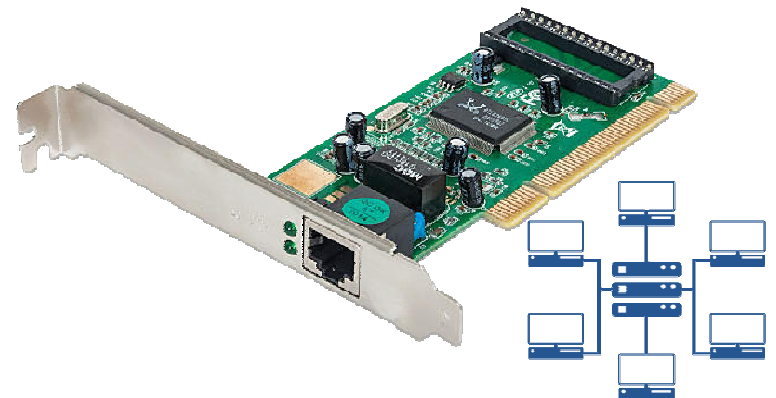
Wireless Media



Wired media

Wired Media:

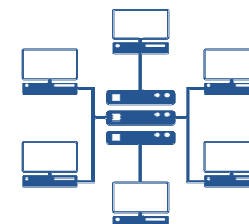
- A physical path exists to have communication between the devices. It is also called as guided media. It uses cables as the medium.
- Ex: Co-axial Cable, Twisted pair cable, etc.
- Wired media Used NIC card.



Wireless media

Wireless Media:

- No physical path exists to have communication over a network. It is also called unguided media. But data is carried using signals like signals like infrared rays, Radio signals, etc.
- Wireless media used Wi-Fi Transmitter and Receiver.

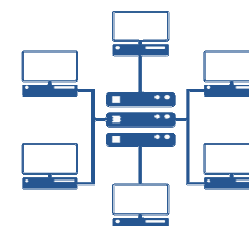


NETWORK DEVICES

Networking devices:

- If a network with **three or more devices wants to communicate over the network** then networking devices (connecting devices) are used to connect two or more devices.

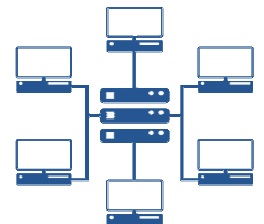
Ex: Hub, Switch, Wireless Router etc.



Protocols

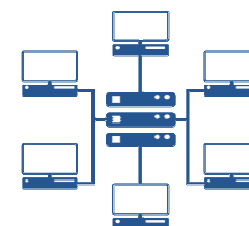
Protocols: Set of rules

- A network **protocol** is a language that is used by systems that wish to communicate with one another.
- If two systems wish to communicate (or talk) with one another, they need to speak the same language (or protocol).

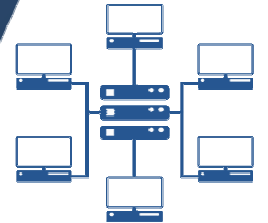
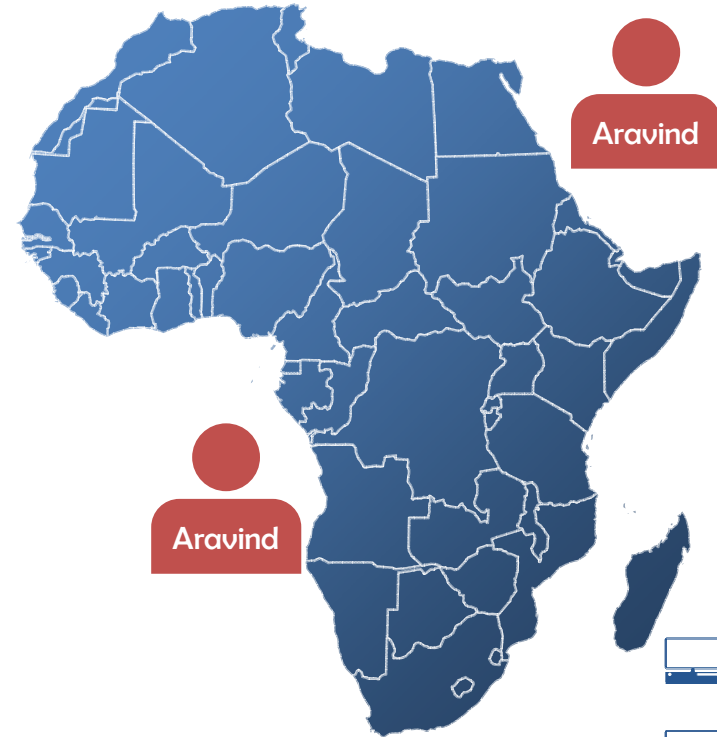
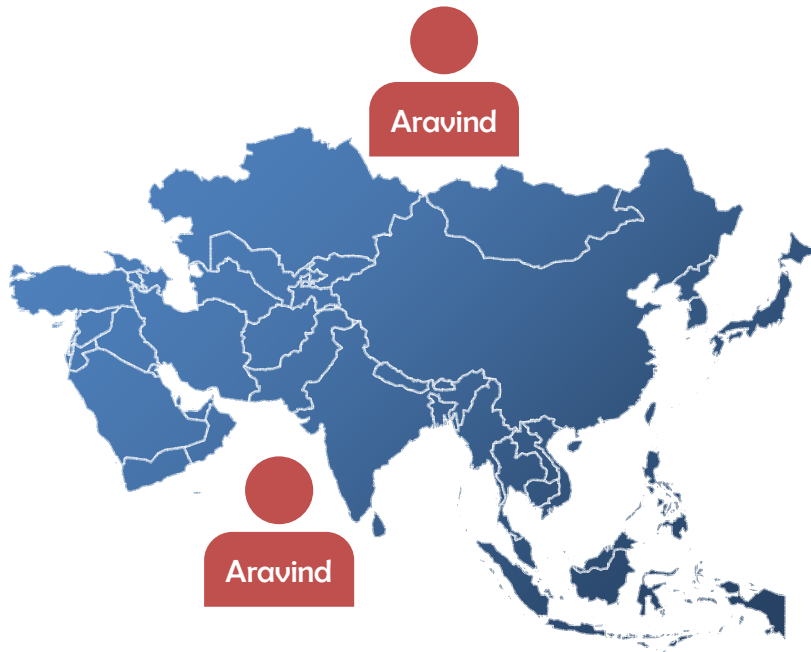


IP Address

- An *Internet Protocol* address (IP address) is a numerical label assigned to each device connected to a computer network that uses the *Internet Protocol* for communication.
- This address is known as logical address.
- This address is used to find the location of the computer.
- IP address can be changed.



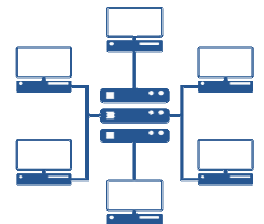
Location-IP Address, Identity-MAC



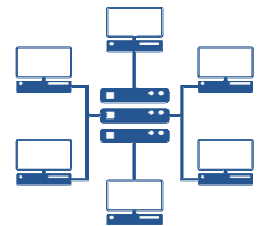
Components required for establishing network



- ✓ Network Enabled Device-Computer/Laptop
- ✓ Media (Wired or Wireless)
- ✓ Networking devices
- ✓ Protocol
- ✓ IP Address



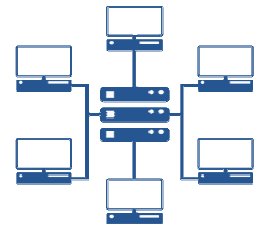
Types of Computer Network Connection?



Types of Network Connections?



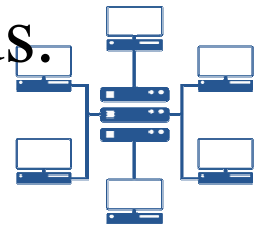
- A network is two or more devices connected through links.
- A link is a communications pathway that transfers data from one device to another.
- For communication to occur, two devices must be connected in some way to the same link at the same time.
- There are two possible types of computer network
 - Point-to-Point(P2P)
 - Multipoint



Point-to-point connection

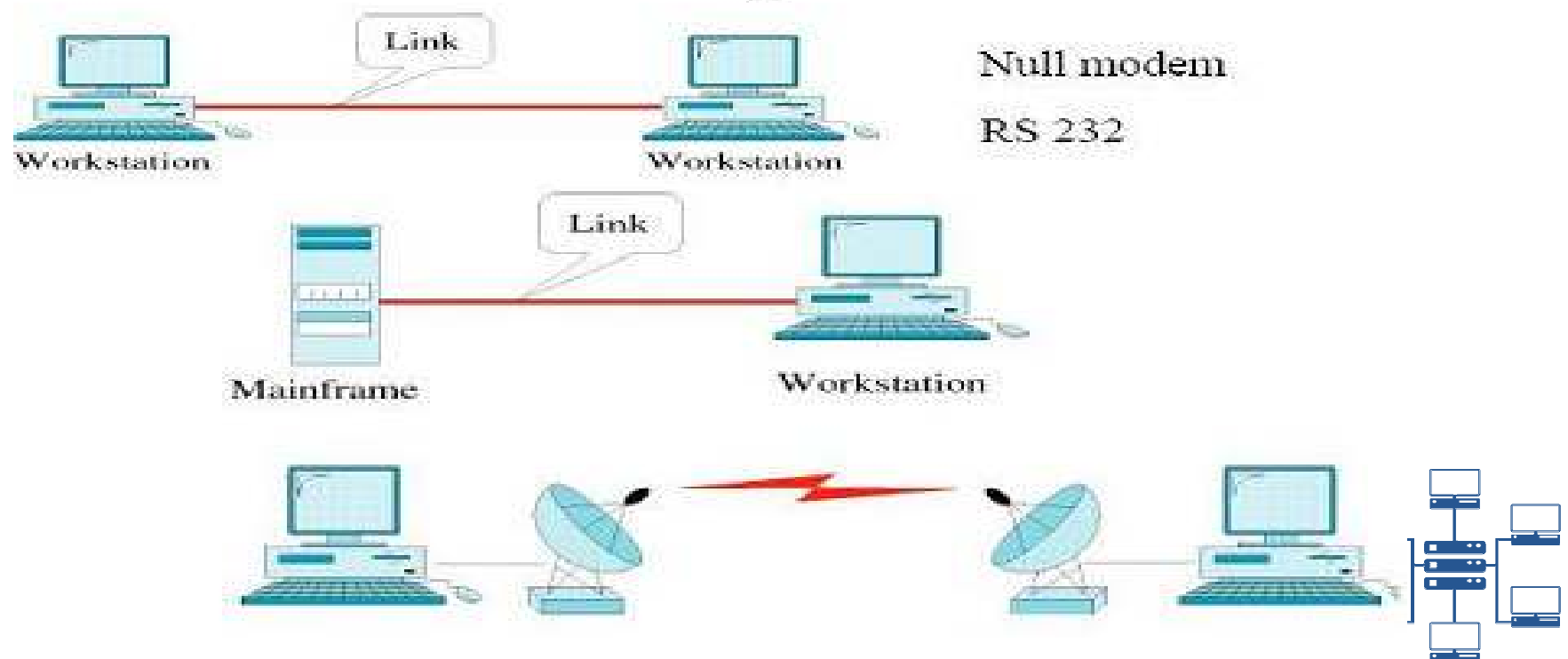


- A point-to-point connection provides a dedicated link between two devices.
- It is also known as Peer to Peer connection.
- The entire capacity of the link is reserved for transmission between those two devices.
- Most point-to-point connections use an actual length of wire or cable to connect the two ends.



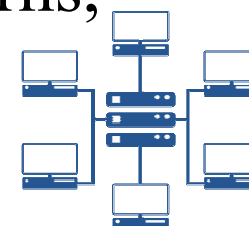
Point to Point connection

Point-to-point

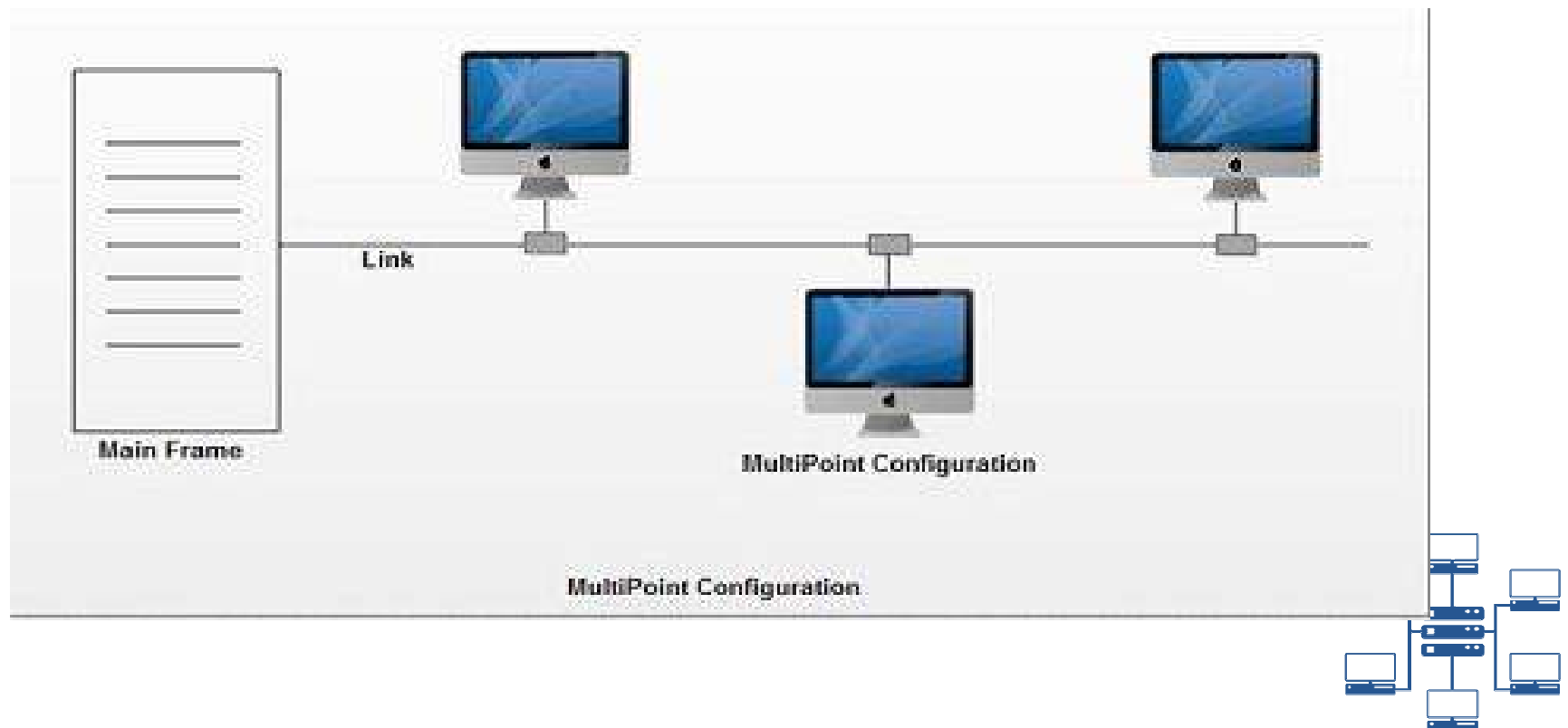


Multipoint connection

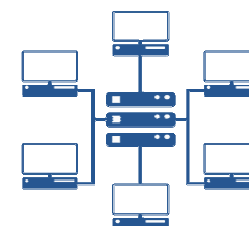
- A multipoint connection is one **in which more than two specific devices share a single link.**
- In a multipoint environment, the **capacity of the channel is shared.**
- Multi-point connection is also known as **client/server network.**
- If several devices can use the link simultaneously, it is a spatially shared connection. If users must take turns, it is a timeshared connection.



Multi Point Connection



How To Share The Data Between Two Computers?

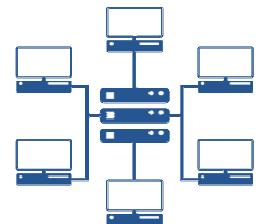


LAB



Remember requirements to connect network.

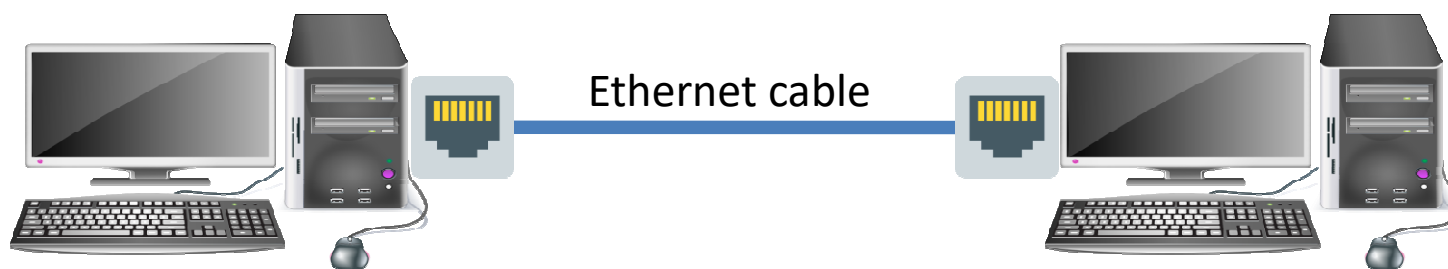
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- ✓ Protocol
- ✓ IP Address



LAB

Remember requirements to connect network.

Point to Point:(Wired)

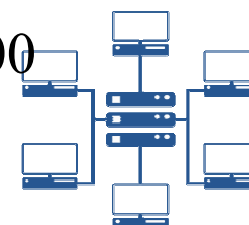


MAC:

IP Address:192.168.1.100

MAC:

IP Address:192.168.1.200



How to Assign an IP address?



- It can be done by two methods.

Manual-Static IP address

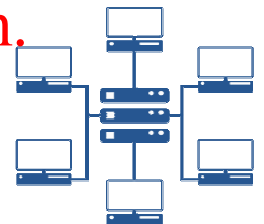
Automatic-Dynamic IP address(DHCP)

Static IP address

A static IP address, the address does not change until we change.

Dynamic IP address

A dynamic IP address is an automatically configured IP address assigned by a DHCP server to every System.
It can be done by modem and router.

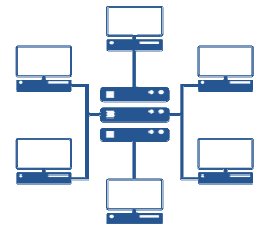


How to Assign IP Address Manually? ?

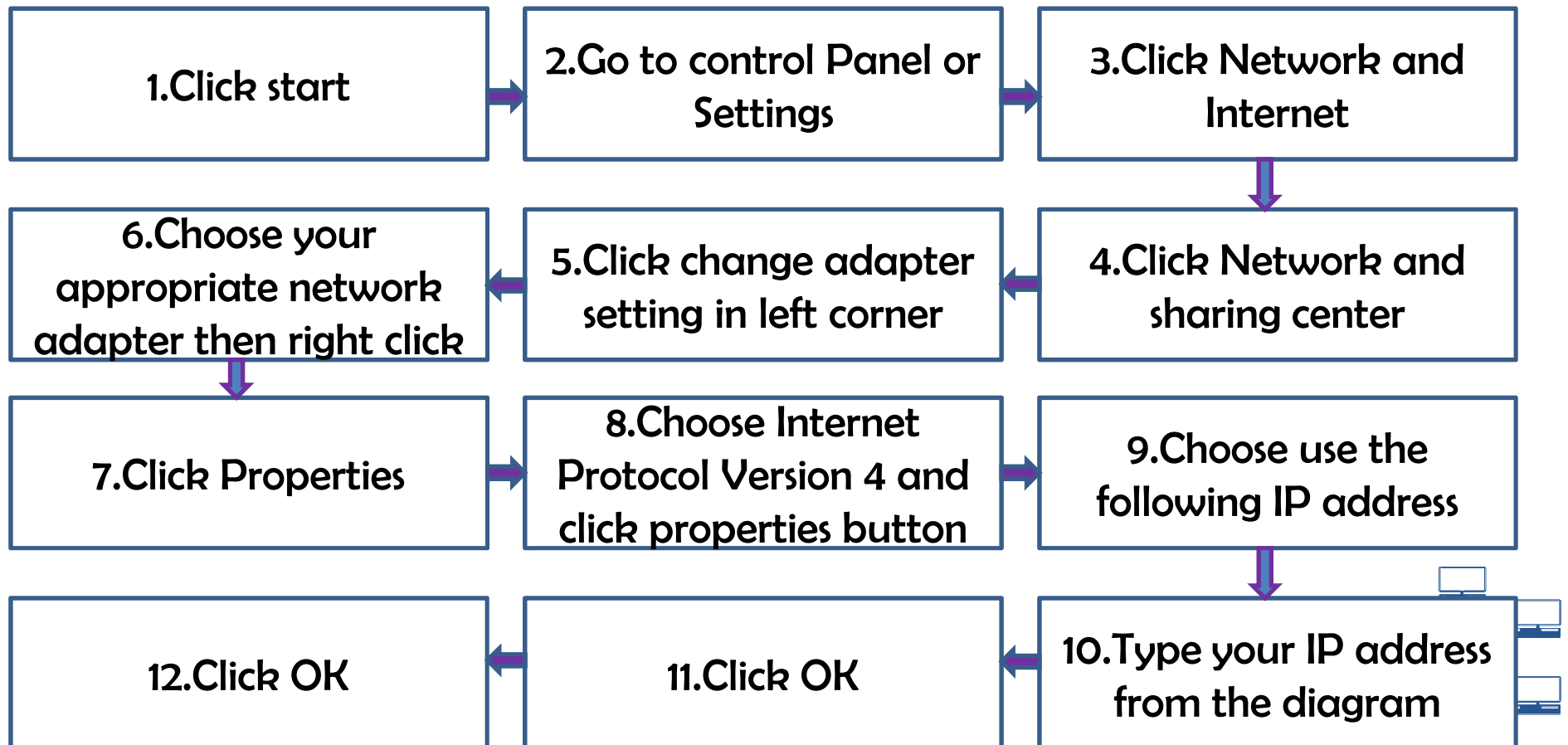
192.168.1.100



192.168.1.200



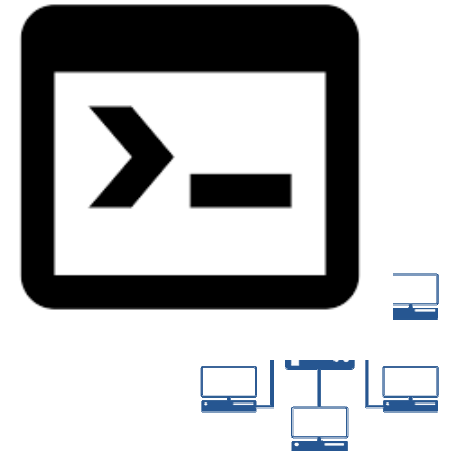
How to Assign Static IP Address?



How to Check an IP Address?



- After assigned an IP address we should check whether **IP address is assigned or not.**
- It **can done by GUI or Command prompt**
- By using command prompt (win+R)
ipconfig

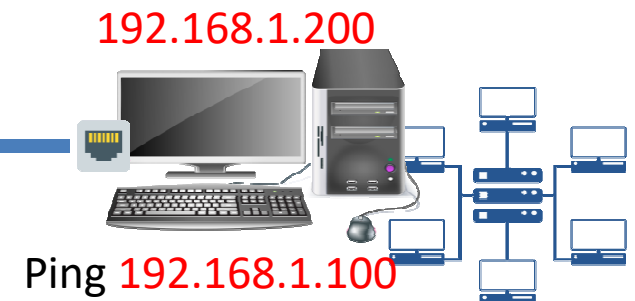


How to Check connectivity Between devices?



- After assigned an IP address we should check connectivity between two devices.
- It **can done by Command prompt**
- By using command prompt

ping <source-ip address>



192.168.1.100



Ping 192.168.1.200

192.168.1.200



Ping 192.168.1.100

```
C:\Users\Aravindraja>ping 192.168.1.200
```

```
Pinging 192.168.1.200 with 32 bytes of data:  
Reply from 192.168.1.200: bytes=32 time=1ms TTL=128  
Reply from 192.168.1.200: bytes=32 time=1ms TTL=128  
Reply from 192.168.1.200: bytes=32 time=1ms TTL=128  
Reply from 192.168.1.200: bytes=32 time=1ms TTL=128
```

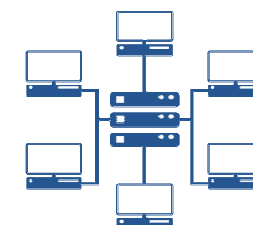
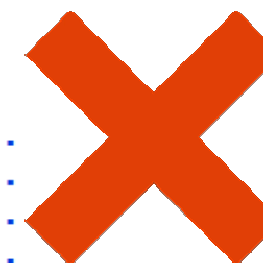


```
C:\Users\Aravindraja>ping 192.168.1.100
```

```
Pinging 192.168.1.100 with 32 bytes of data:  
Reply from 192.168.1.100: bytes=32 time=1ms TTL=128  
Reply from 192.168.1.100: bytes=32 time=1ms TTL=128  
Reply from 192.168.1.100: bytes=32 time=1ms TTL=128  
Reply from 192.168.1.100: bytes=32 time=1ms TTL=128
```

```
C:\Users\Aravindraja>ping 192.168.1.100
```

```
Pinging 192.168.1.100 with 32 bytes of data:  
Reply from 192.168.1.200: Destination host unreachable.  
Reply from 192.168.1.200: Destination host unreachable.  
Reply from 192.168.1.200: Destination host unreachable.  
Reply from 192.168.1.200: Destination host unreachable.
```

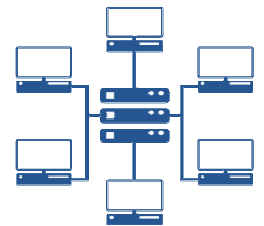


Troubleshooting

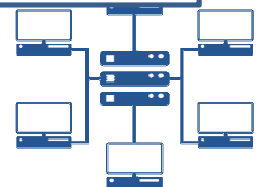
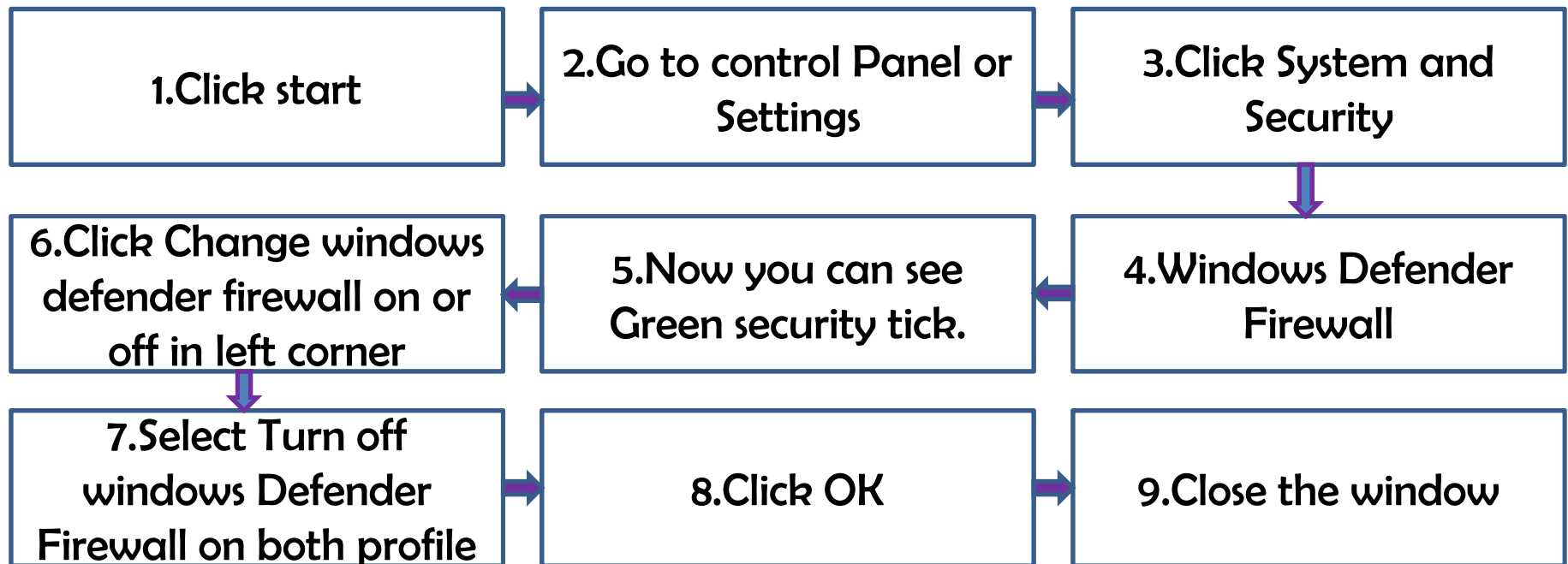


OFF

FIREWALL



How to Turn Off the Firewall?



192.168.1.100



Ping 192.168.1.200

192.168.1.200



Ping 192.168.1.100

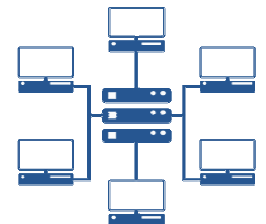


```
C:\Users\Aravindraja>ping 192.168.1.200
```

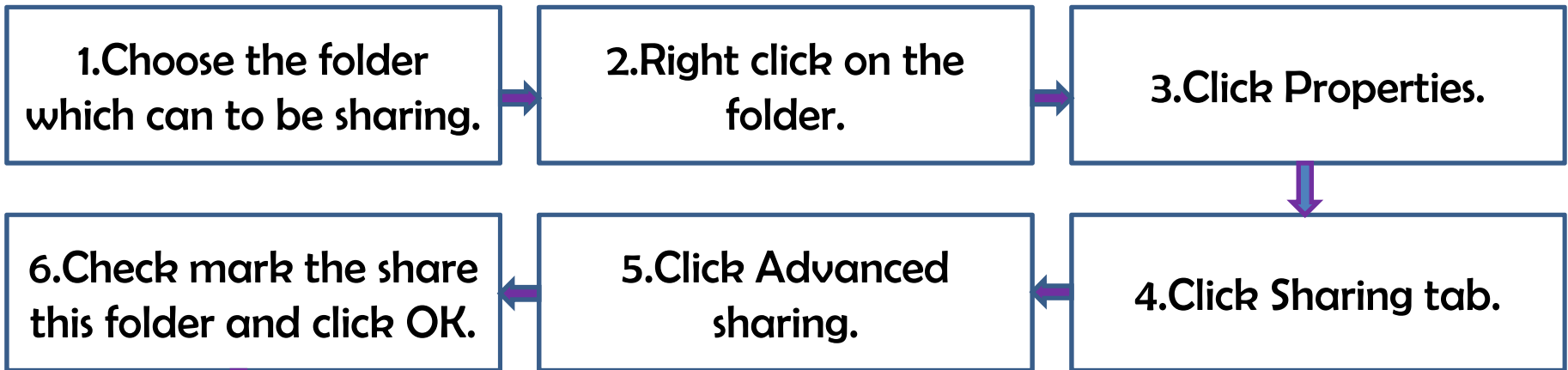
```
Pinging 192.168.1.200 with 32 bytes of data:  
Reply from 192.168.1.200: bytes=32 time=1ms TTL=128  
Reply from 192.168.1.200: bytes=32 time=1ms TTL=128  
Reply from 192.168.1.200: bytes=32 time=1ms TTL=128  
Reply from 192.168.1.200: bytes=32 time=1ms TTL=128
```

```
C:\Users\Aravindraja>ping 192.168.1.100
```

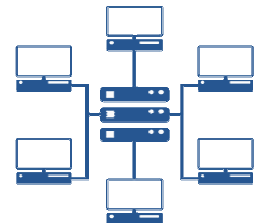
```
Pinging 192.168.1.100 with 32 bytes of data:  
Reply from 192.168.1.100: bytes=32 time=1ms TTL=128  
Reply from 192.168.1.100: bytes=32 time=1ms TTL=128  
Reply from 192.168.1.100: bytes=32 time=1ms TTL=128  
Reply from 192.168.1.100: bytes=32 time=1ms TTL=128
```



How to Share the data?



Now we will see the network path to access the data from another computer over the network.



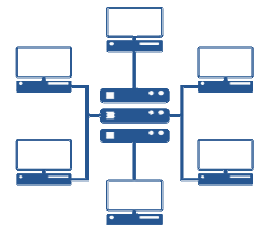
How to Access the shared Folder?



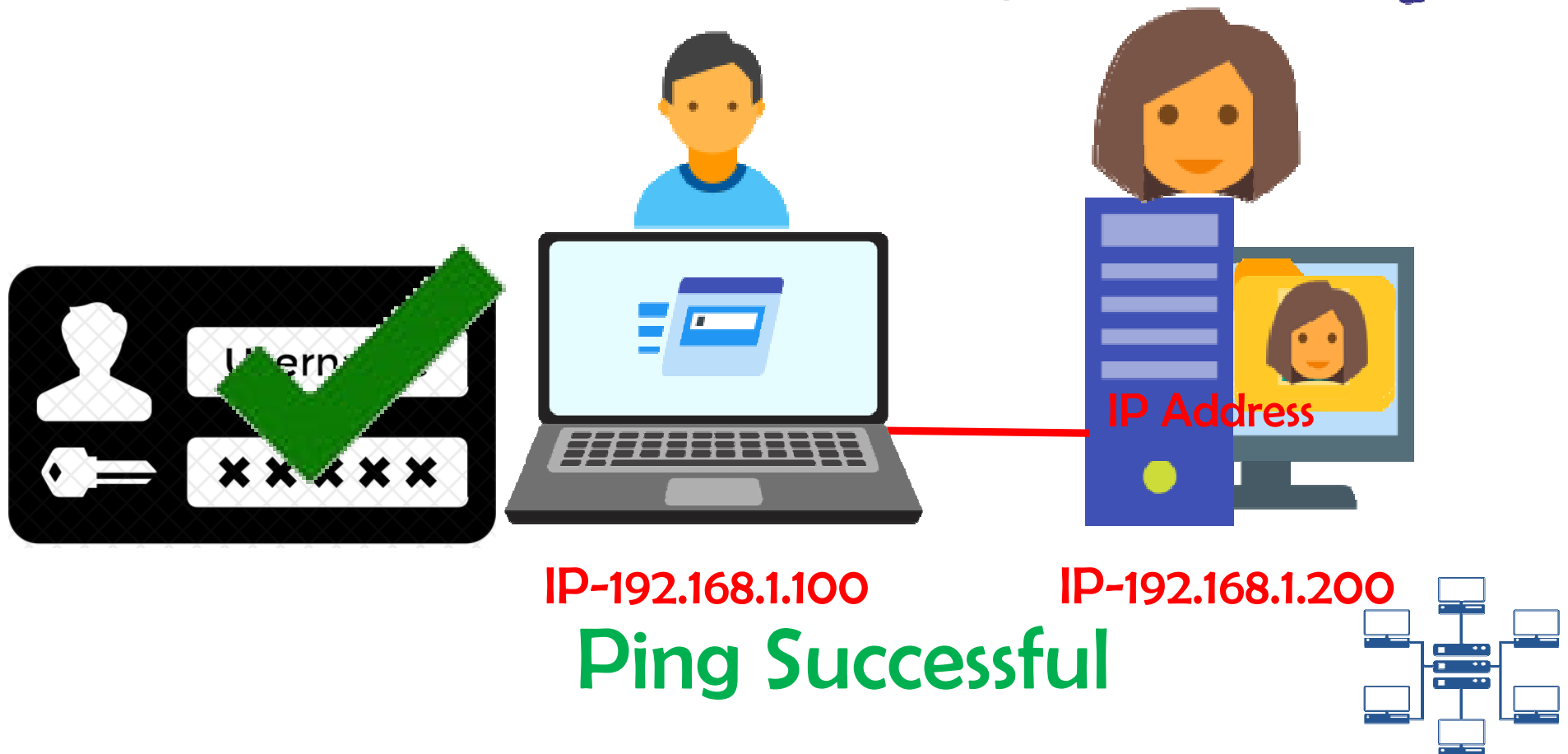
- ✓ It can be done by 'n' no of ways.
- ✓ Open run prompt(press win+r)
- ✓ Type your source ip address like this

Example: UNC path

\\192.168.1.100 or \\network path

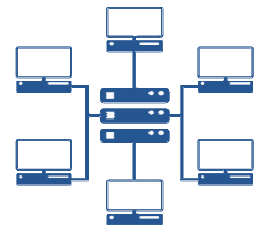


IP Address based Sharing





Multipoint Connection?

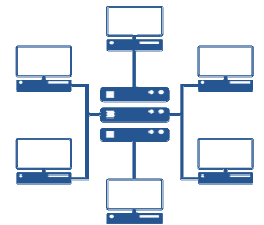


LAB-Multipoint connection Data Sharing



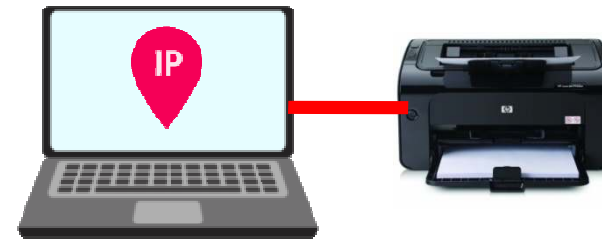
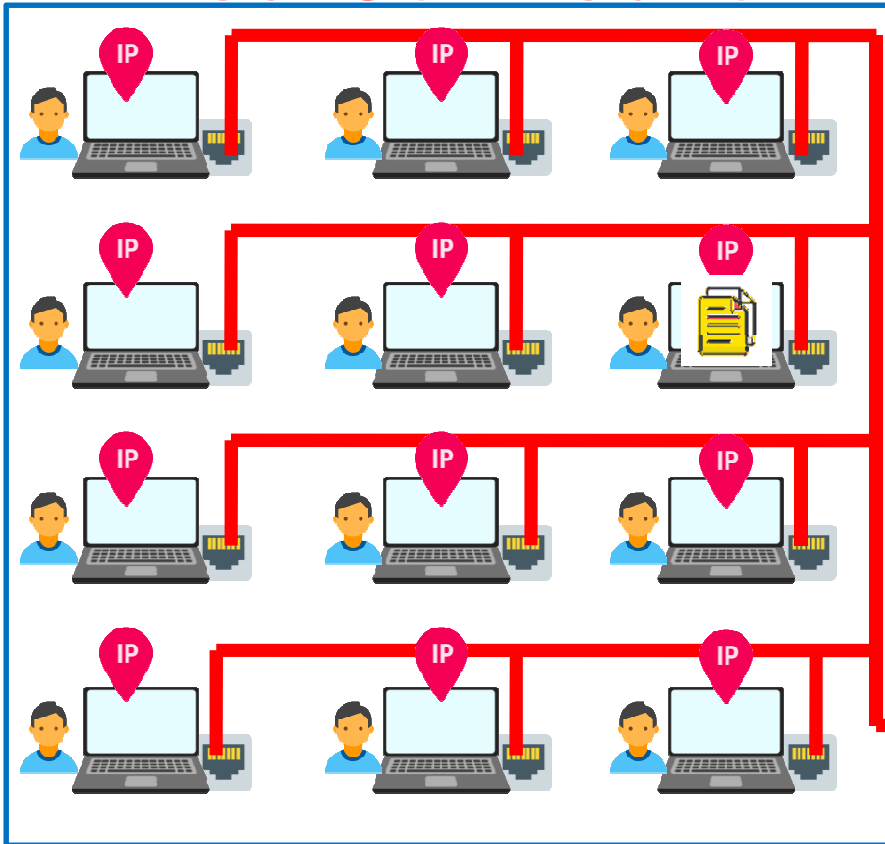
Remember requirements to connect network.

- ✓ Network Enabled Device-Computer/Laptop
- ✓ Media (Wired or Wireless)
- ✓ Networking devices
- ✓ Protocol
- ✓ IP Address

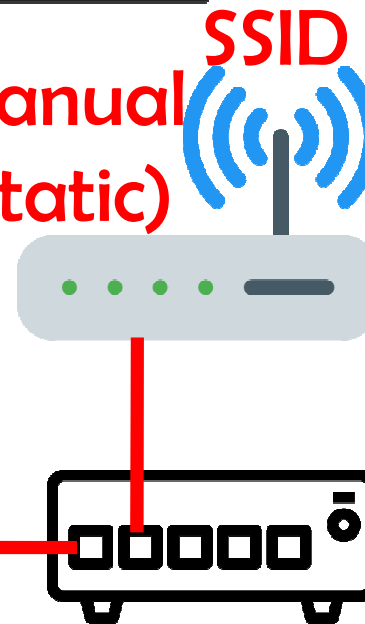


Our Scenario 192.168.1.xxx

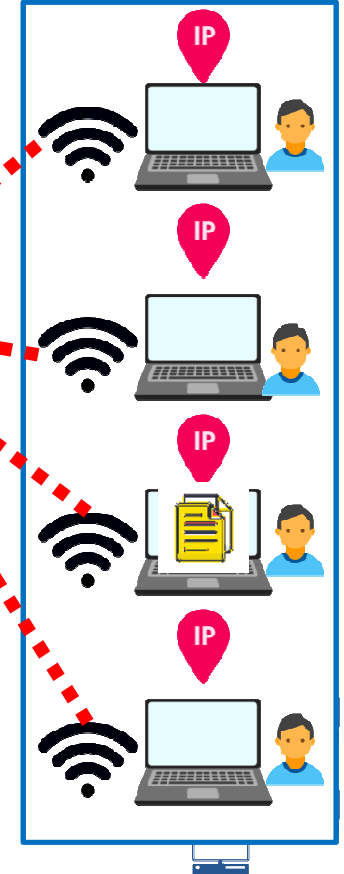
Wired Connection



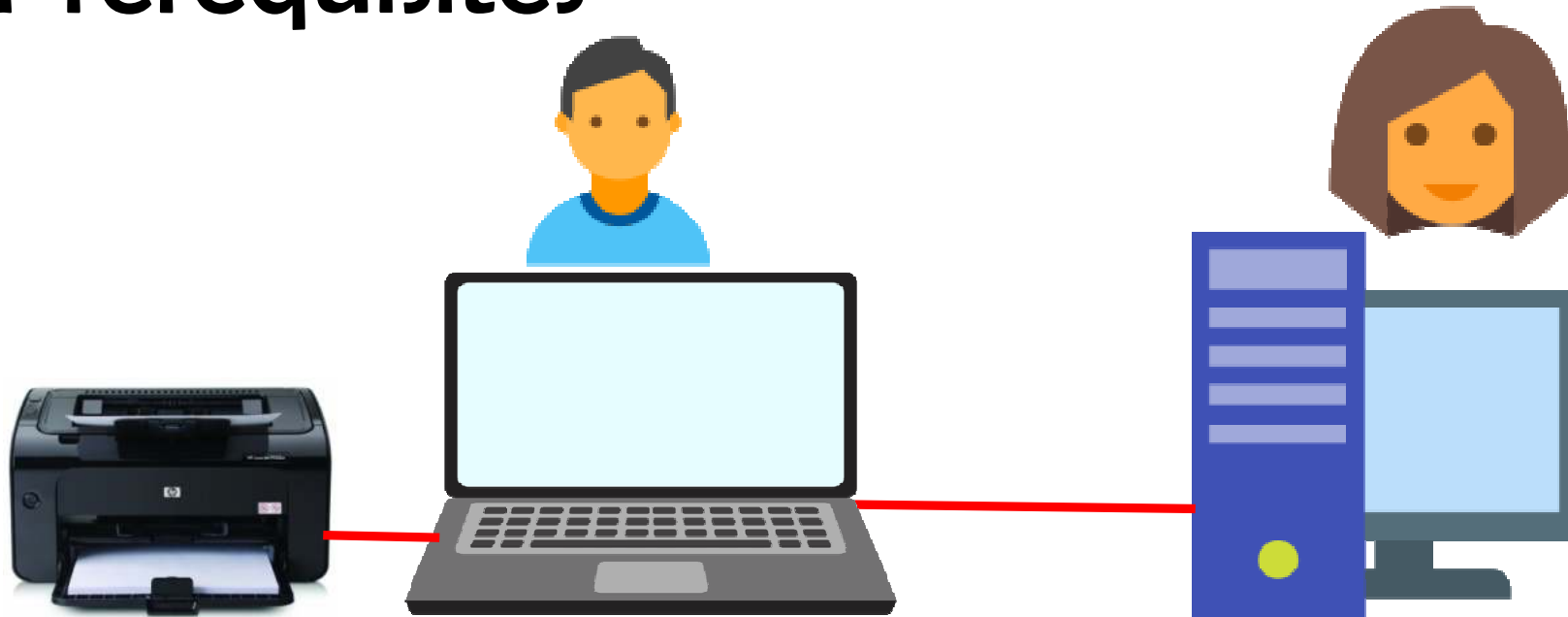
Manual
(Static)



Wireless



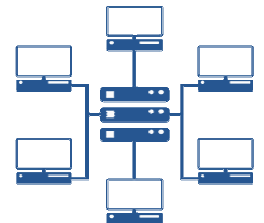
Prerequisites



IP-192.168.1.100

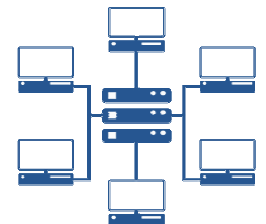
IP-192.168.1.200

Ping Successful

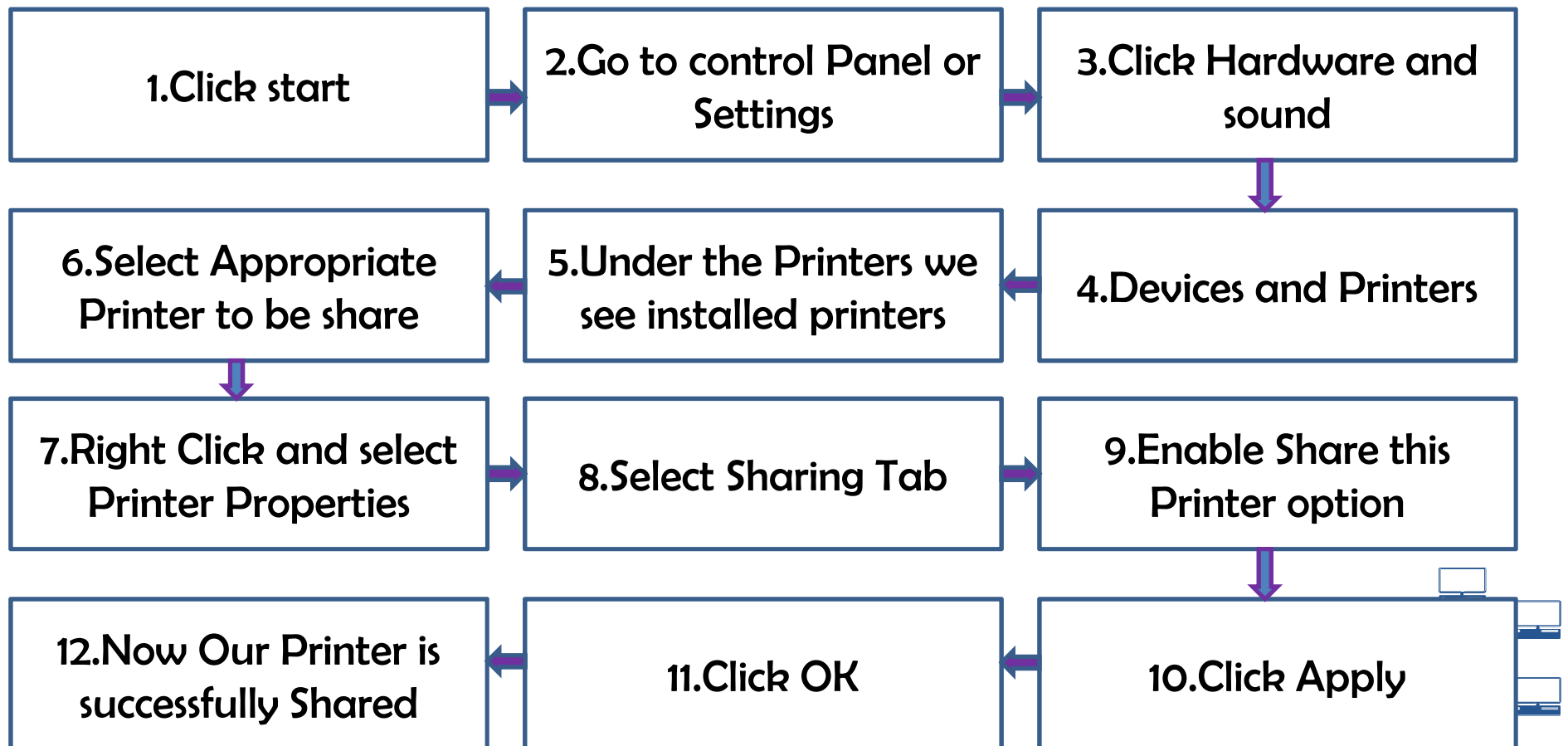


How to Install the Printer?

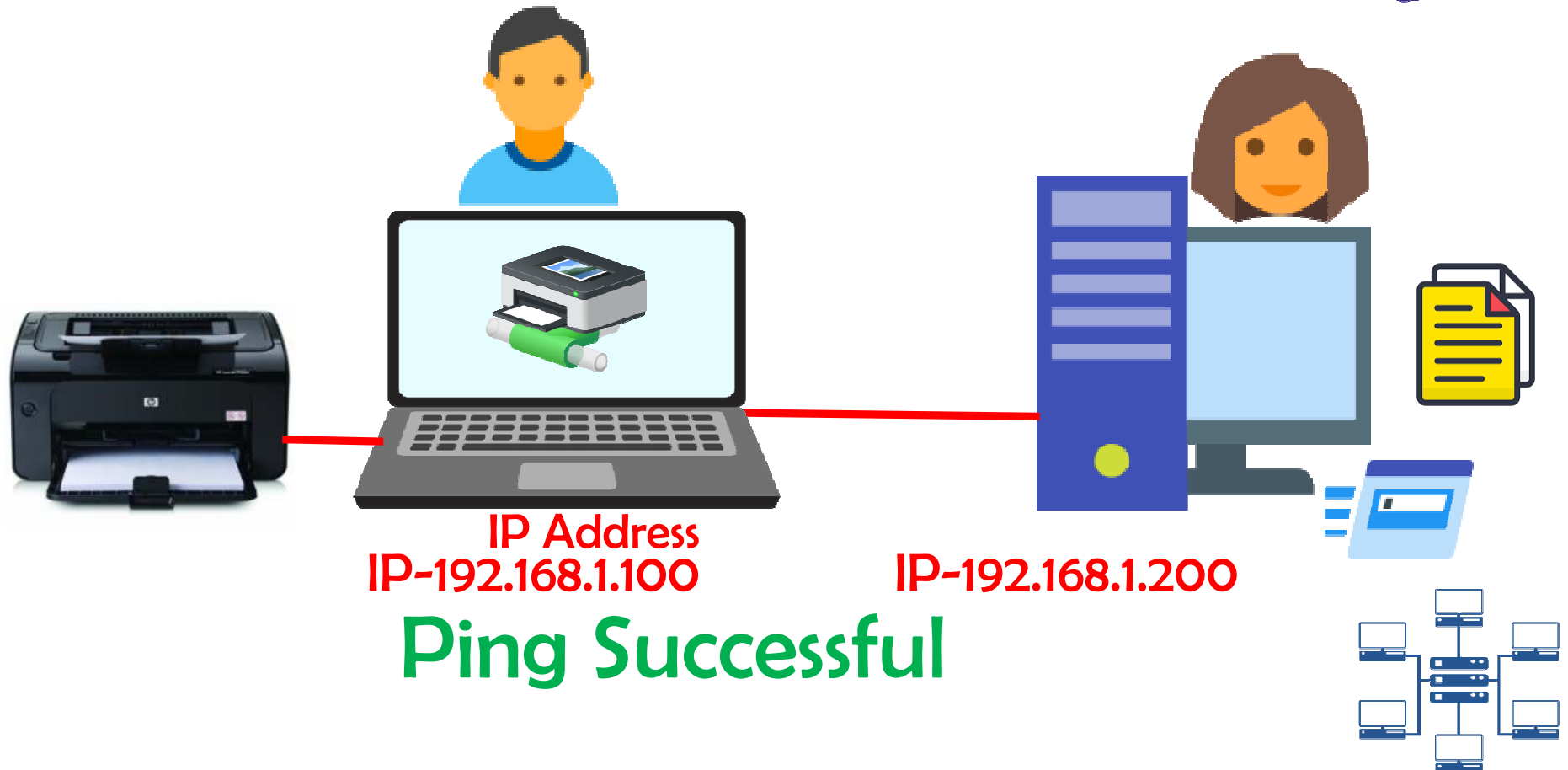
Drivers



How to Share the printer in Physical Machine?



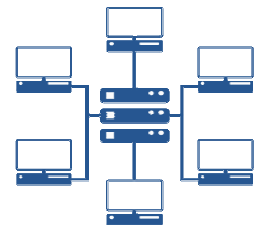
How to Access the Shared Printer?



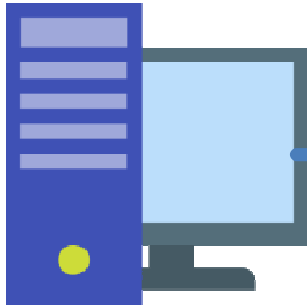
How to access the shared printer?



- ✓ It can be done by 'n' no of ways.
- ✓ It can done by same way as how to access shared data.
- ✓ Type your source ip address(where the printer installed in physical machine).
- ✓ You will see the shared printer icon and double click it(to install printer).
- ✓ Then check your physical machine on printers and drivers option, now printer shared successfully.
- ✓ Now give a print to the printer.



Question?



MAC:AA

IP:192.168.1.100

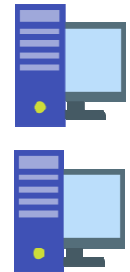
Ping 192.168.1.200



MAC:BB

IP:192.168.1.200

Ping 192.168.1.10



Pinging 192.168.1.200 with 32 bytes of data:

Reply from 192.168.1.200 : bytes=32 time=9ms TTL=64

Reply from 192.168.1.200 : bytes=32 time=8ms TTL=64

Reply from 192.168.1.200 : bytes=32 time=3ms TTL=64

Reply from 192.168.1.200 : bytes=32 time=4ms TTL=64

Ping statistics for 192.168.1.200 :

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 3ms, Maximum = 9ms, Average = 6ms



C:\Users\Aravindraja>ping 192.168.1.10

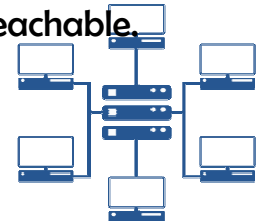
Pinging 192.168.1.10 with 32 bytes of data:

Reply from 192.168.1.10: Destination host unreachable.

Reply from 192.168.1.10: Destination host unreachable.

Reply from 192.168.1.10: Destination host unreachable.

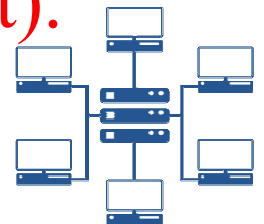
Reply from 192.168.1.10: Destination host unreachable.



ICMP Protocol?



- ✓ ICMP-Internet Control Message Protocol.
- ✓ It gives success or failure message from source to destination.
- ✓ Ping is a utility which uses ICMP message to report back information on network connectivity between host and a destination computer.
- ✓ Every segment or ICMP payload must be encapsulated within an IP datagram (or packet).

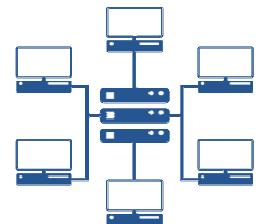


ARP Protocol?



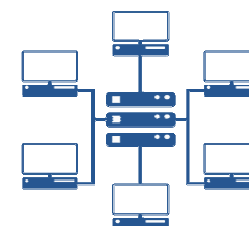
ARP Protocol:

- ✓ ARP-Address Resolution Protocol.
- ✓ ARP is a network protocol used to find out the hardware(MAC) address of a device from an IP address.
- ✓ It is used to resolve IP to MAC address.
- ✓ It is done by the ARP table.



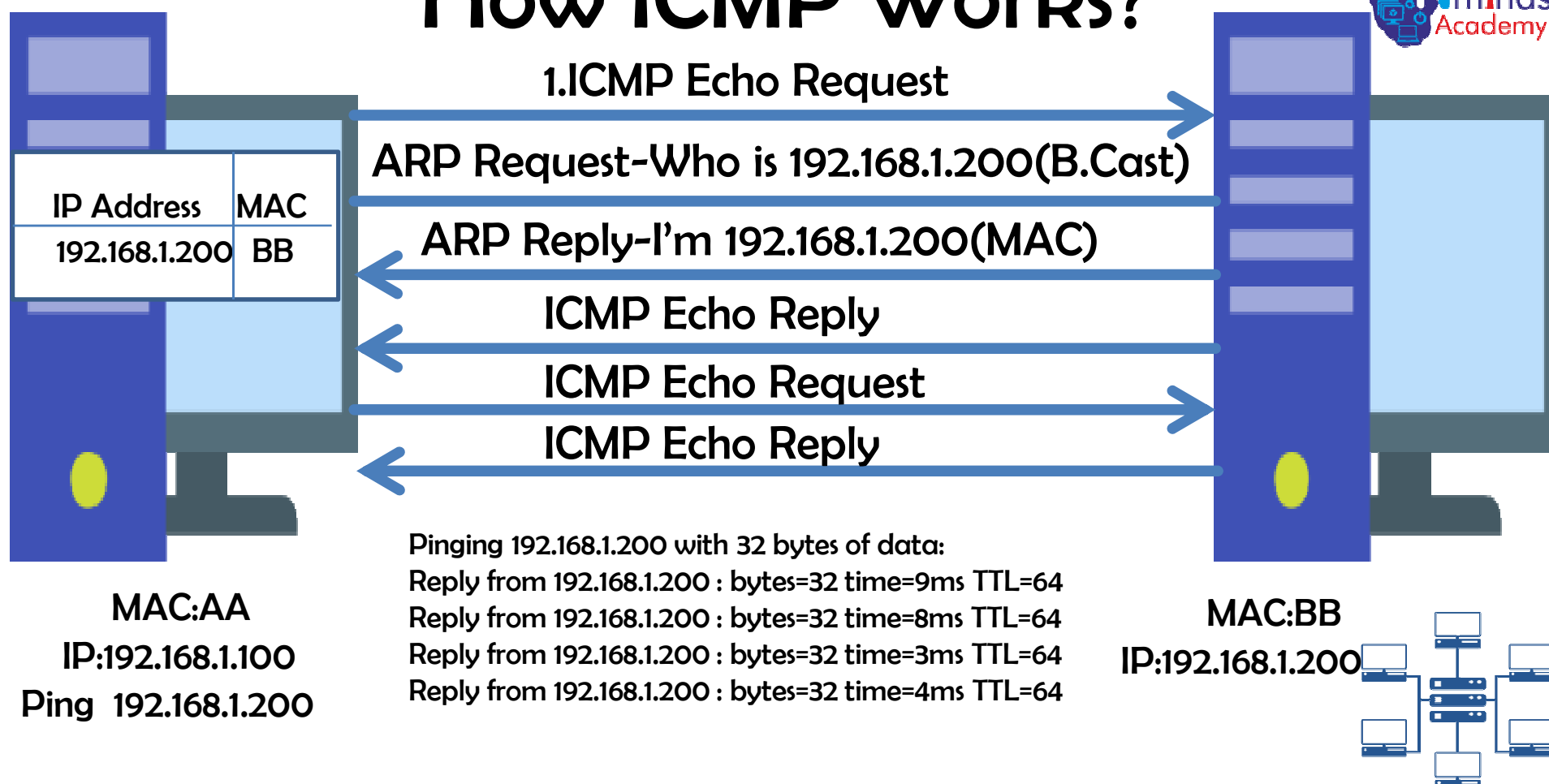
ARP Table?

- ✓ **Address Resolution Protocol (ARP)** is the method for finding a host's Link Layer (MAC) address when only its IP address is known.
- ✓ The ARP table is **used to maintain a correlation between each MAC address and its corresponding IP address.**
- ✓ We can see this ARP table by using command
`arp -a`



How ICMP Works?

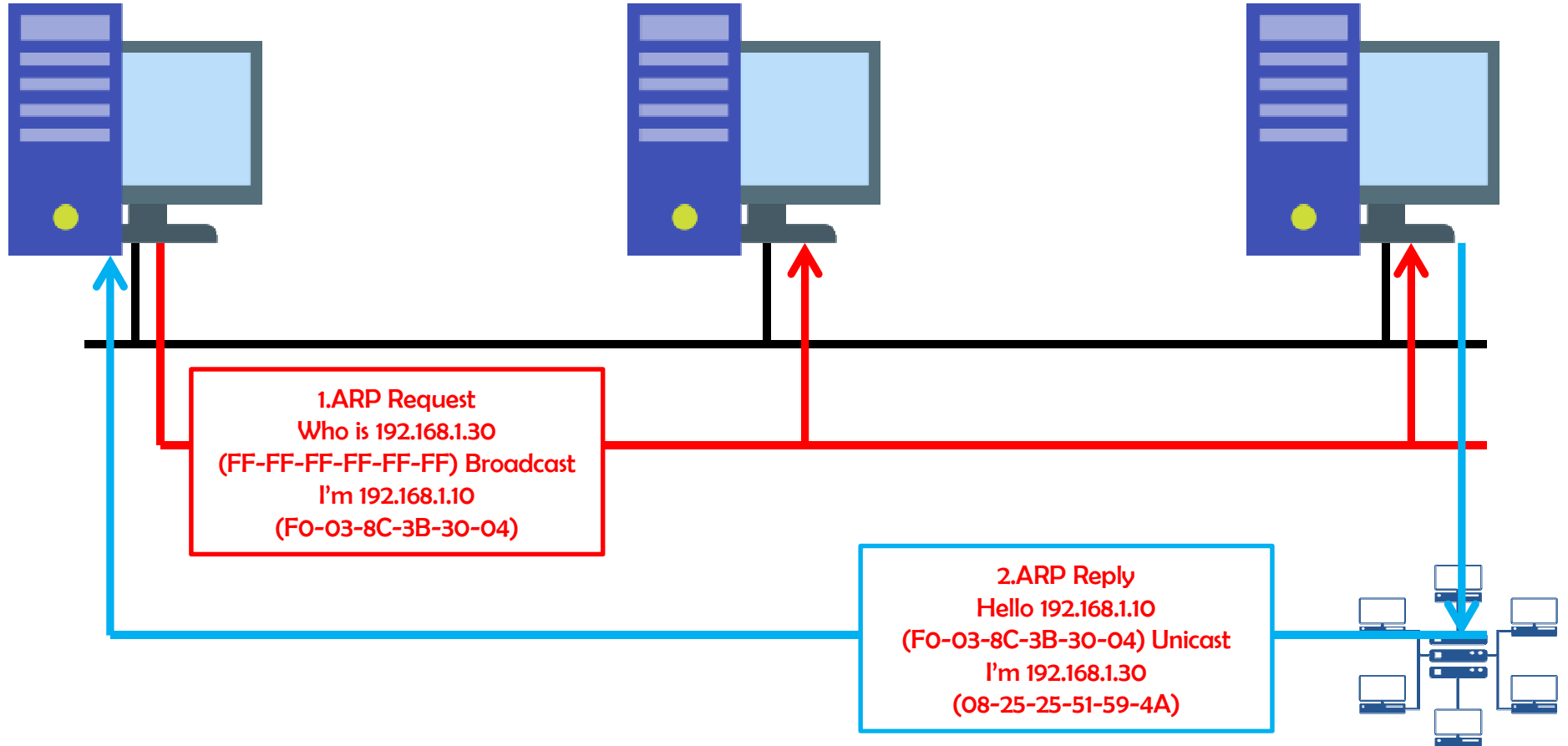
1.ICMP Echo Request



MAC: Fo-03-8C-3B-30-04
IP: 192.168.1.10
Ping 192.168.1.30

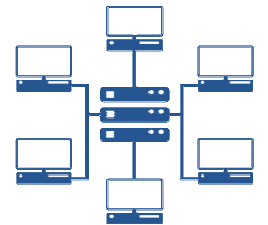
MAC: 18-60-24-48-B1-37
IP: 192.168.1.20

MAC: 08-25-25-51-59-4A
IP: 192.168.1.30





How to build Network?



How to build network?

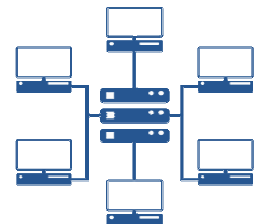


- ✓ It can be done by types of network.
- ✓ In order to build a network we have different types.

Based on Distance

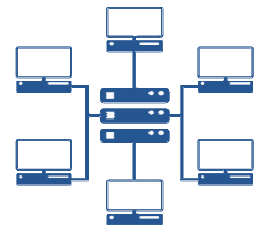
Based on Architecture

Based on Connection



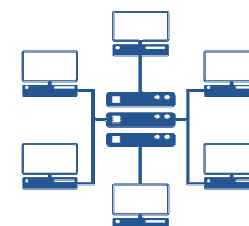


BASED ON DISTANCE



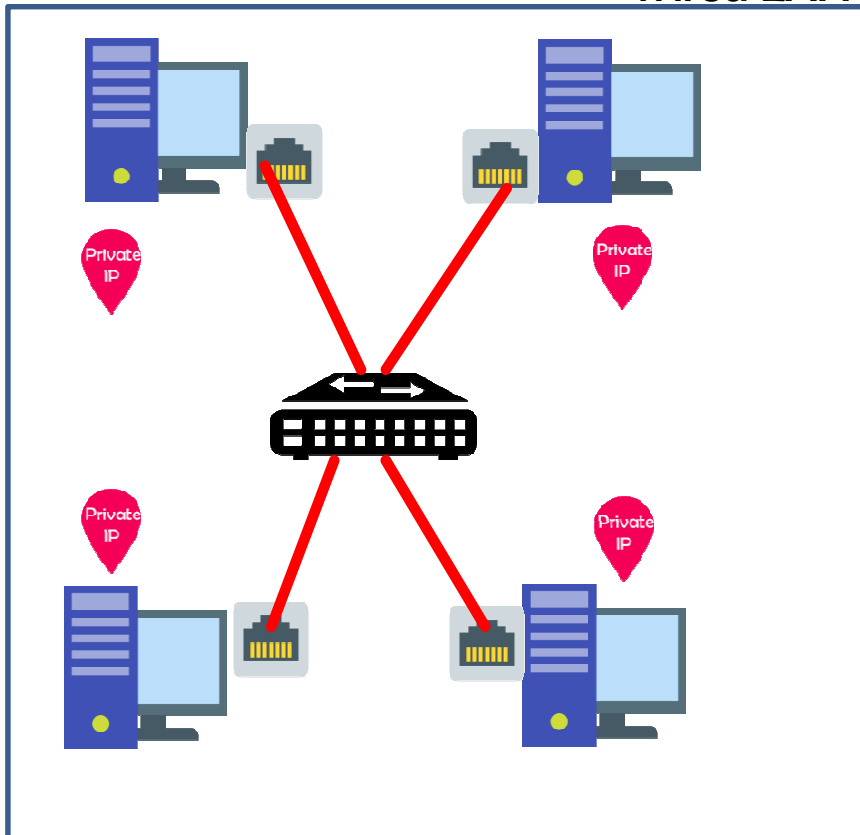
Types of network

- ✓ This types of **network is based on distance.**
- ✓ LAN-Local Area Network
- ✓ MAN-Metropolitan Area Network
- ✓ WAN-Wide Area Network
- ✓ CAN-Campus Area Network
- ✓ PAN-Personal Area Network
- ✓ SAN-Storage Area Network

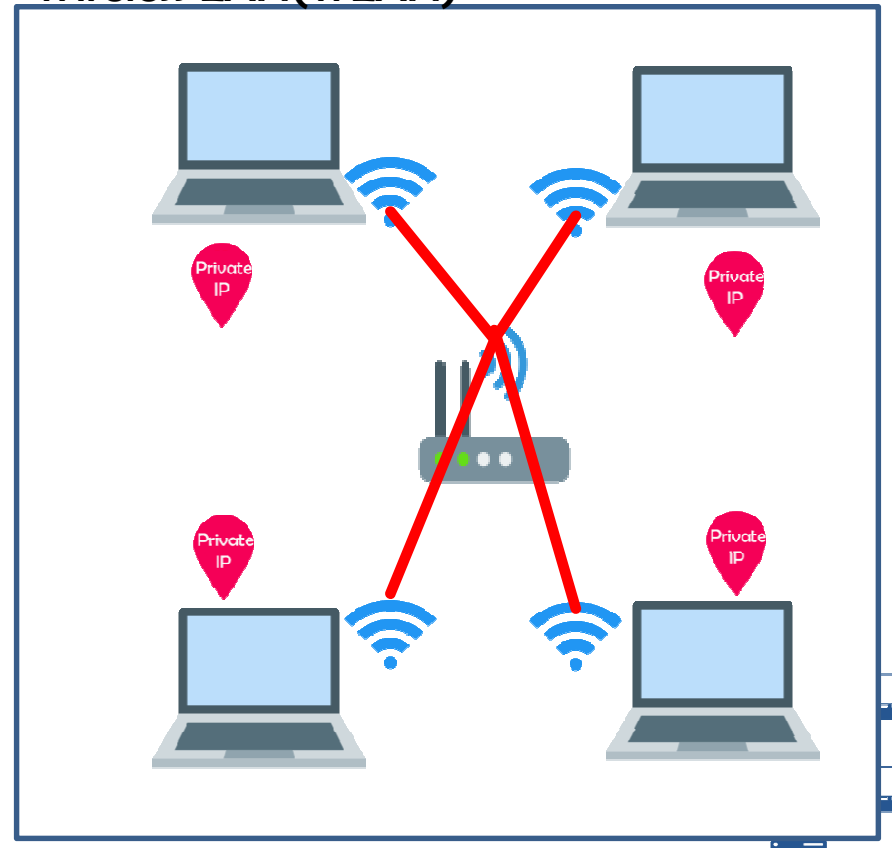


LAN-How it Work's?

Wired LAN



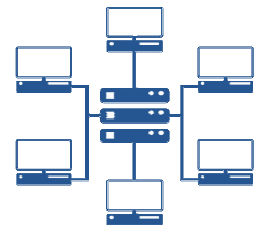
Wireless LAN(WLAN)



LAN



- ✓ LAN stands for **Local Area Network**.
- ✓ A LAN is a **group of computers and devices that are in a specific location**.
- ✓ **LAN** computers can **exchange data and messages in the easy and fast way**. It also saves time and makes our work fast.
- ✓ The **devices connect to the LAN with an Ethernet cable(NIC) or through Wi-Fi(Wi-Fi card)**.
- ✓ It **not requires internet connection**.
- ✓ It **uses Private IP address for Internal communication Purpose, Given** by Network Administrator.
- ✓ LAN used by School,college,company....



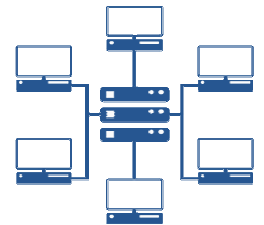
Advantages and disadvantages of LAN?

Advantages

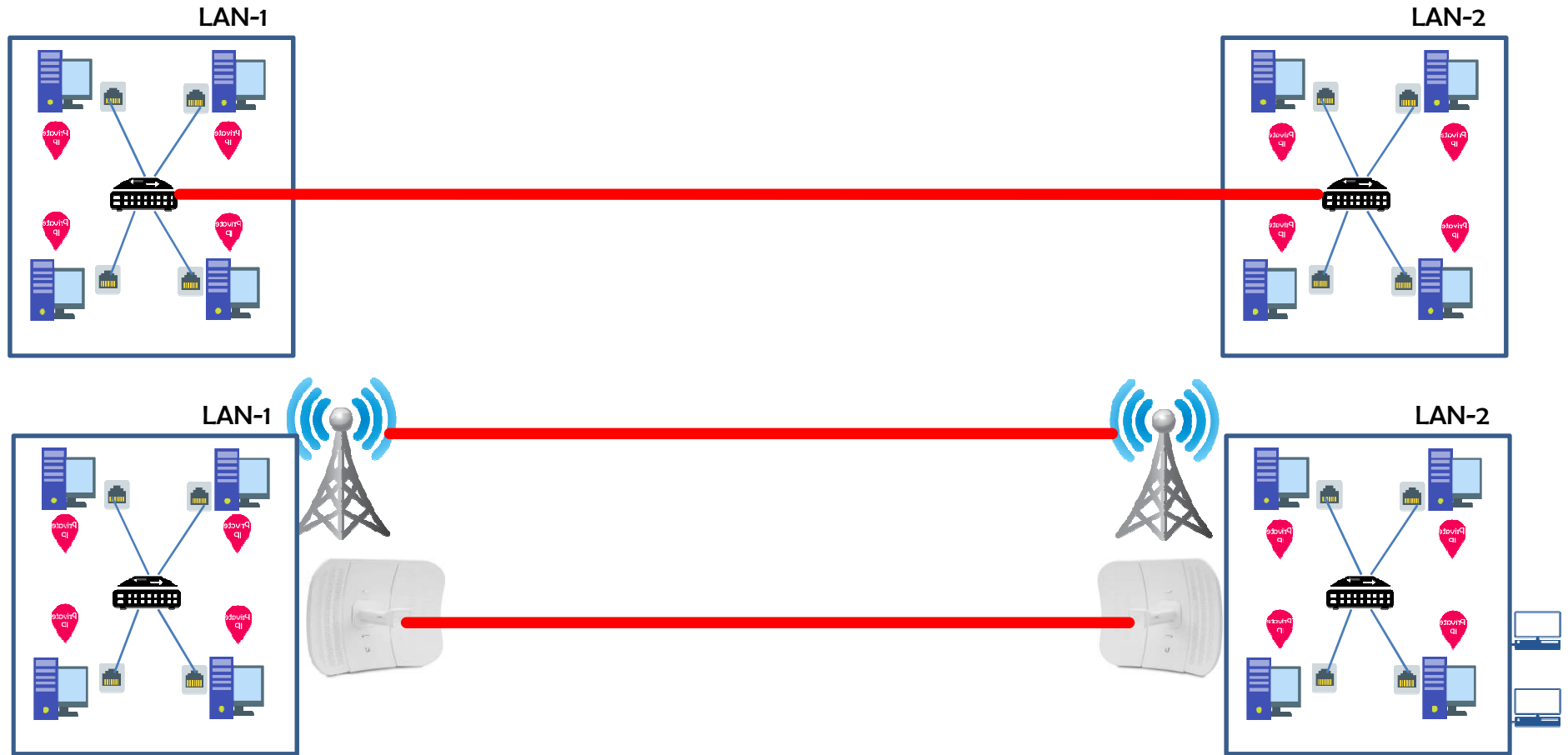
- ✓ Easy to share devices
- ✓ Data sharing is easy and with easy performance
- ✓ The cost of creating a local area network is low.
- ✓ Creating a complex network might give a user a lot of experience.

Disadvantage

- ✓ Security isn't always 100% ensured.
- ✓ The power must be on at all times to make LAN work efficiently.
- ✓ Upkeep is hard to maintain on regular basis.



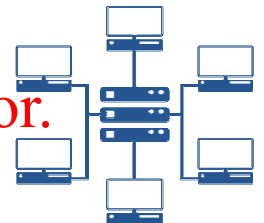
MAN-How it Work's?



MAN



- ✓ MAN stands for **Metropolitan Area Network**.
- ✓ A metropolitan area network is a computer network that **interconnects users with computer resources in a geographic region of the size** of a metropolitan area.
- ✓ **Interconnection of two or more than two LAN's with geographic region without internet connection.**
- ✓ The **devices connect to the LAN with an Ethernet cable(Switch) or through Wi-Fi(RF).**
- ✓ It is also used to mean the interconnection of several local area networks by bridging them with backbone lines.
- ✓ It **not requires Internet connection.**
- ✓ It **uses Private IP address, Given by Network administrator.**
- ✓ Use:school,College,company.....



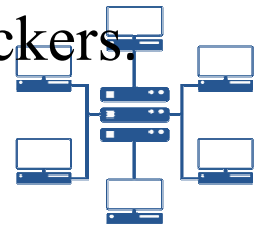
Advantages and disadvantages of MAN

Advantages

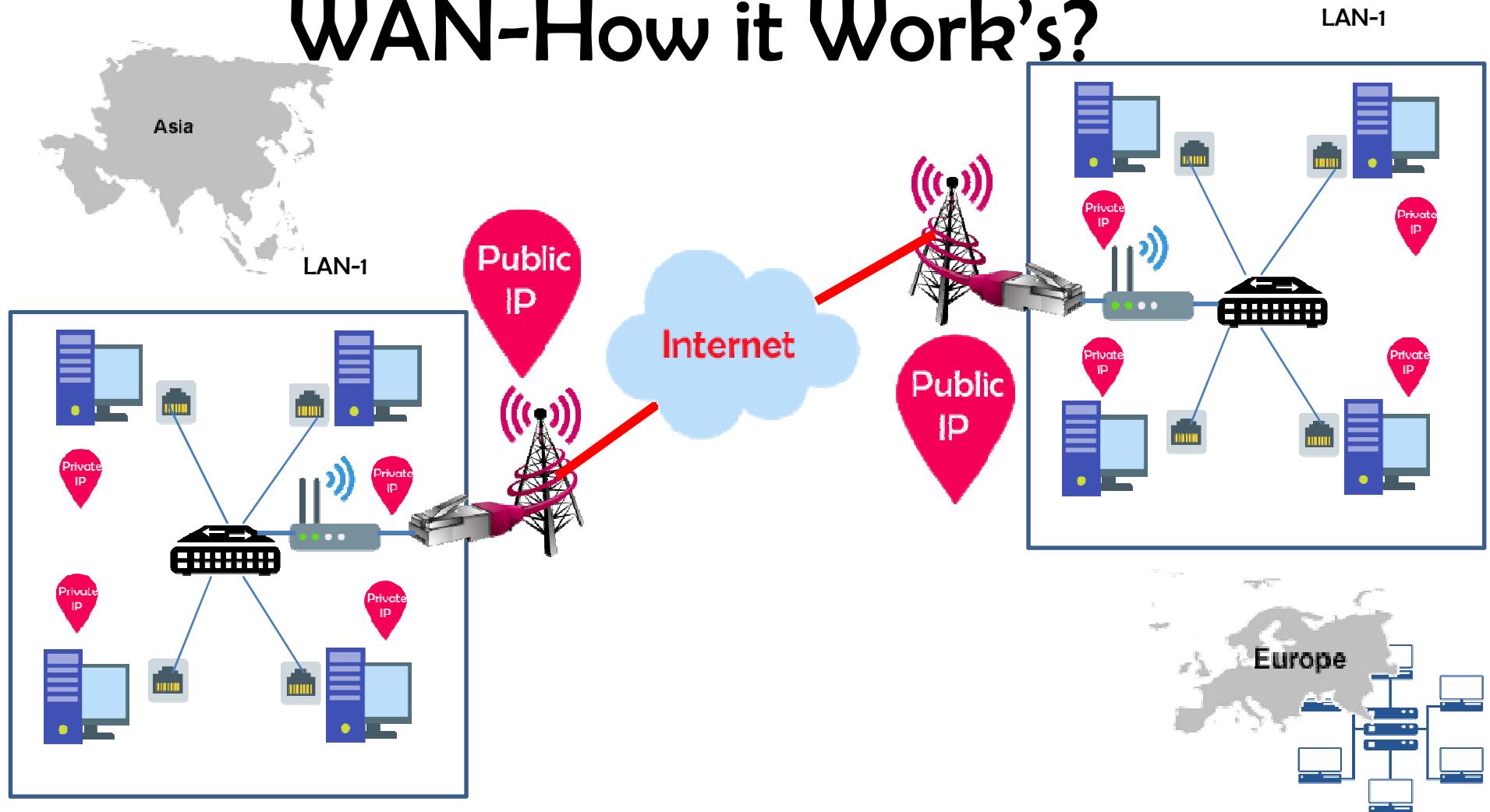
- ✓ It provides a good back bone for a large network.
- ✓ The dual bus used in MAN helps the transmission of data in both direction simultaneously.
- ✓ Enable people to connect many fast LANs together.
- ✓ Cost effective by sharing management of data.

Disadvantage

- ✓ The large network becomes difficult to manage.
- ✓ More cable required for a MAN connection from one place to another.
- ✓ It is difficult to make the system secure from hackers.



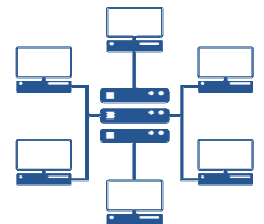
WAN-How it Work's?



WAN



- ✓ WAN stands for Wide Area Network.
- ✓ A **WAN** connects different smaller **networks**, including local area **networks** (LANs) and metro area **networks** (MANs) in different geographical area.
- ✓ This ensures that computers and users in **one location can communicate with computers and users in other locations by using internet connection.**
- ✓ It **requires Internet connection.**
- ✓ **Internet connection provided by ISP**(Internet service provider)
- ✓ It **uses public IP address for external communication.**
- ✓ **This Public IP is given by ISP.**



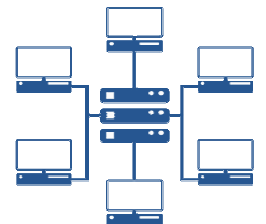
Advantages and disadvantages of WAN

Advantages

- ✓ Covers a large geographical area(Business can connect long distance).
- ✓ Allows connecting workstation to share resource and software.
- ✓ Message can be sent very quickly to anyone else on the network.

Disadvantage

- ✓ High costs.
- ✓ Low security
- ✓ To improve our security to avoid disruption by hackers.
- ✓ Once set up, maintaining a network is a full-time job which requires network supervisors and technicians to be employed.



Comparison of LAN & WAN

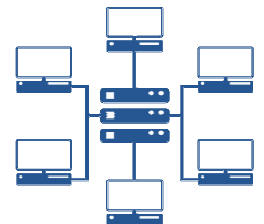


LAN

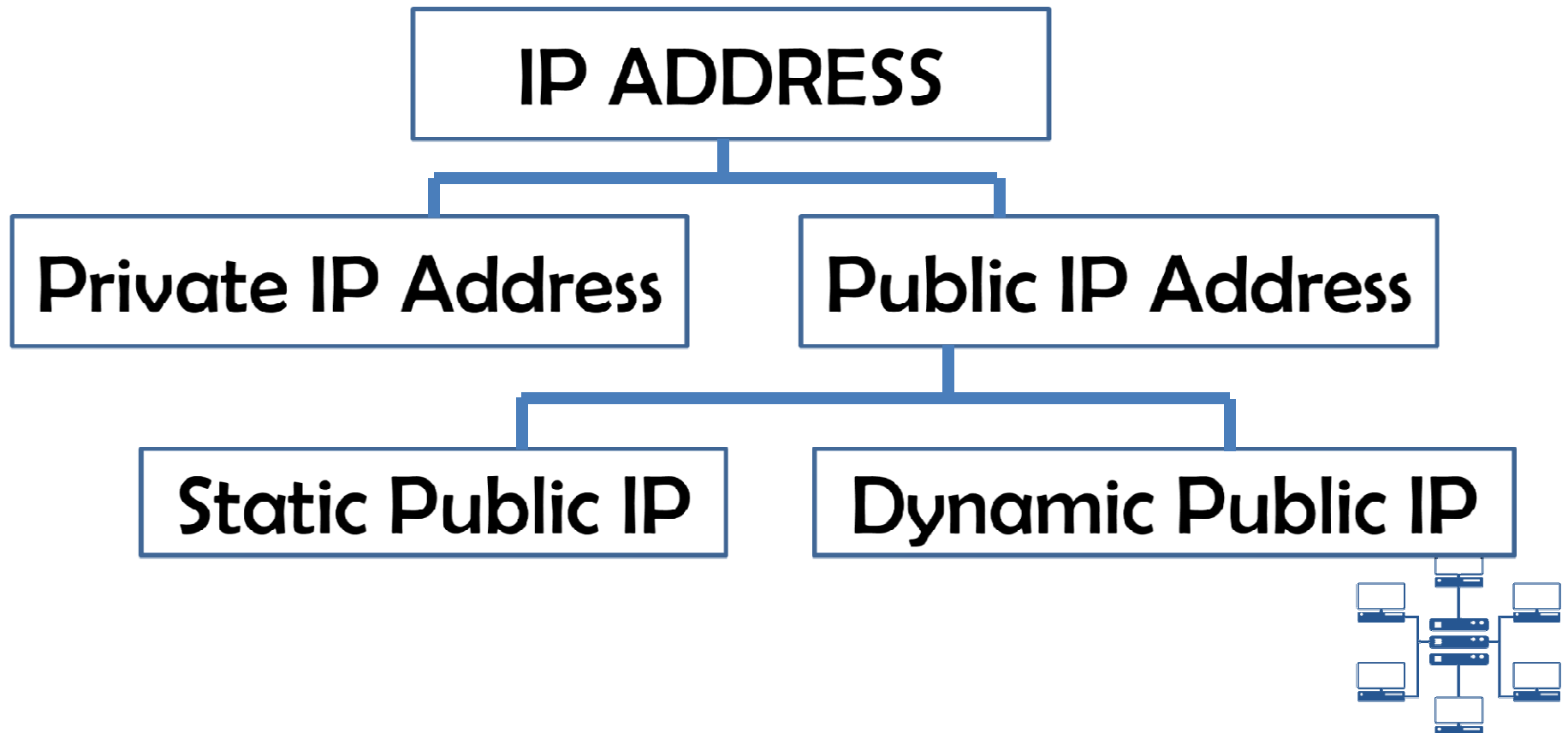
- ✓ Local Area Network
- ✓ Within Office/Same LAN
- ✓ It not requires internet
- ✓ It uses Private IP address
- ✓ It free of cost
- ✓ IP given by Network Administrator

WAN

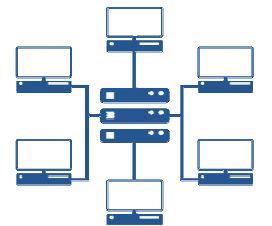
- ✓ Wide Area Network
- ✓ Connect different LAN
- ✓ It requires Internet
- ✓ It uses Public IP address
- ✓ It is cost
- ✓ IP given by Internet Service Provider(ISP)



Types of IP Address



Protocol

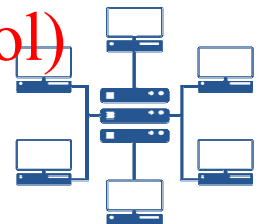


PROTOCOLS



Protocols: **Set of rules**

- ✓ Protocol -a **set of rules or procedure for transmitting data between electronic devices, such as computers.**
- ✓ A network protocol is a language that is used by systems that **wish to communicate with one another.**
- ✓ In order for computers to exchange information, there **must be a pre-existing agreement as to how the information will be structured and how each side will send and receive it.**
- ✓ If two **systems wish to communicate (or talk) with one another, they need to speak the same language(protocol)**



PROTOCOLS



SMB

HTTP

HTTPS

RDP

FTP

TFTP

Telnet

DNS

DHCP

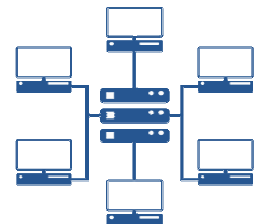
SMTP

IMAP

POP3

SNMP

NTP



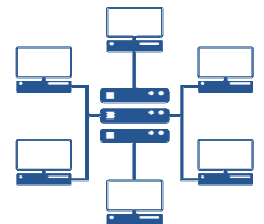
Example



PORT NO



- ✓ A port number is a way to identify a specific process to which an Internet or other network message is to be forwarded when it arrives at a server.
- ✓ A port number uniquely identifies a network-based application on a computer.
- ✓ Port numbers are part of the addressing information used to identify the senders and receivers of messages.
- ✓ Each application/program is allocated a 16-bit integer port number.



PROTOCOLS-Port No

SMB-445

HTTP-80

HTTPS-443

RDP-3389

SSH-22

FTP-21,20

TFTP-69

SMTP-25

Telnet-23

DNS-53

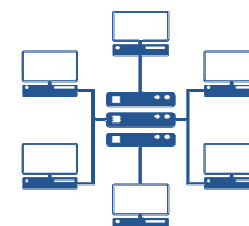
DHCP-67,68

LDAP-389

POP3-110

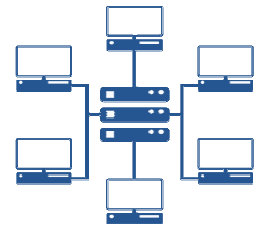
SNMP-161

IMAP-143



SMB

SERVER MESSAGE BLOCK-445,139



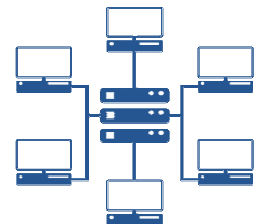
What is SMB Protocol?



- ✓ The **Server Message Block protocol** (SMB protocol) is a client-server communication **protocol used for sharing access to files, printers and other resources on a network.**
- ✓ **SMB** is a network file sharing protocol that requires an open port on a computer or server to communicate with other systems.

IP -445(TCP)

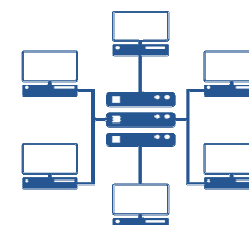
NETBIOS Name-139(TCP)



How to Access the data?

- ✓ **UNC- Universal Naming Convention**
- ✓ This UNC path is used to access the shared data.
- ✓ It uses \\ (backward slash)
- ✓ In a network, the **Universal Naming Convention (UNC)** is a way to identify a shared file in a computer without having to specify (or know) the storage device it is on.

\\ip-address



How to access shared folder?

COMPUTER OR LAPTOP

Application Name:

Windows:

RUN prompt-win+r

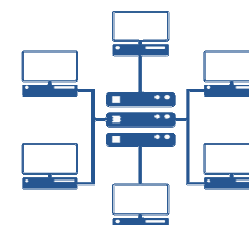


MOBILE PHONE

Application Name:

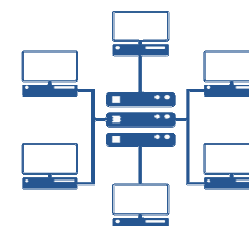
Go To Play store

AndSMB(Samba Client)



RDP

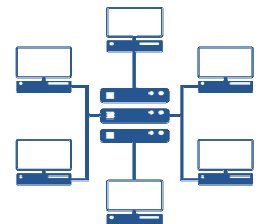
REMOTE DESKTOP PROTOCOL-3389



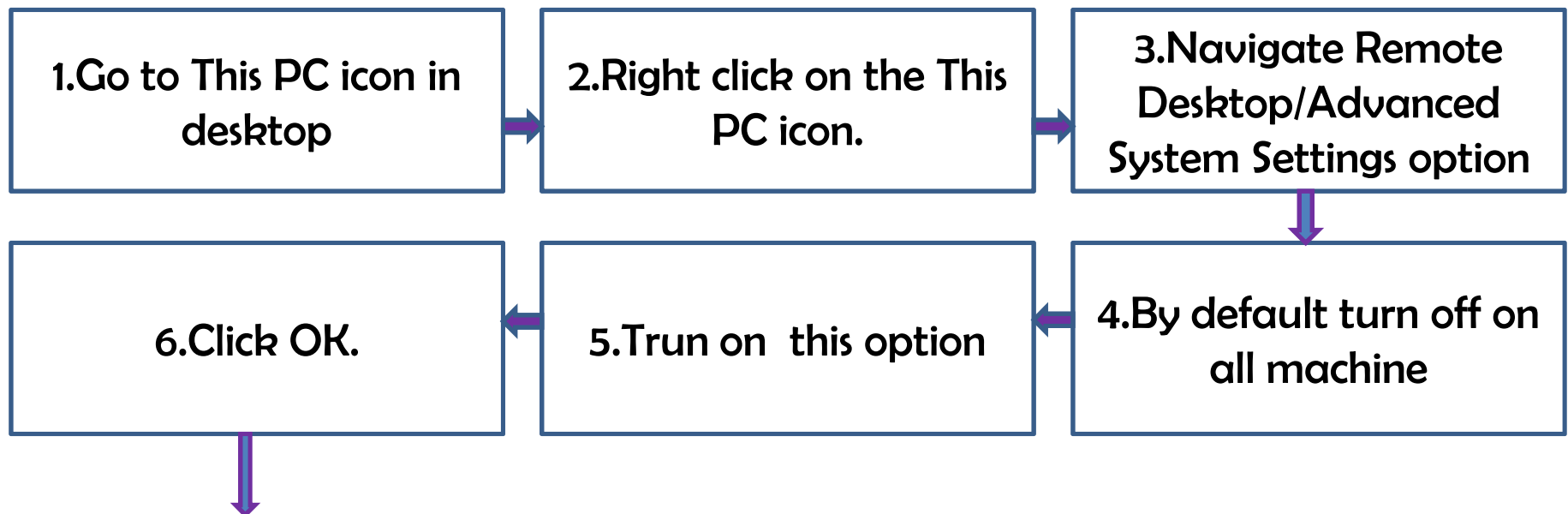
What is RDP Protocol?



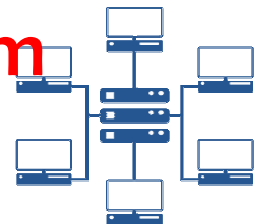
- ✓ **Remote Desktop Protocol (RDP)** is a proprietary protocol developed by Microsoft which provides a user with a **graphical interface to connect to another (remote) computer** over a network connection.
- ✓ This protocol is used for take remote desktop.
- ✓ **RDP** is an extremely popular **protocol** for remote access to Windows machines.
- ✓ Port no is **3389(TCP)**.



How to Configure RDP Protocol?



Now this machine will able to take remotely from another computer over the network.



How to Access RDP Protocol?

COMPUTER OR LAPTOP

Software Name:

Windows:



Remote Desktop Connection

Linux:

Remmina



VNC

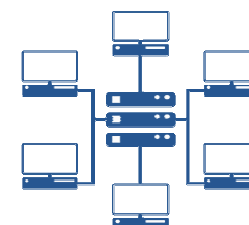


MOBILE PHONE

Software Name:

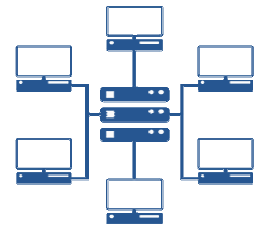
Go To Play store

Remote Desktop



HTTP

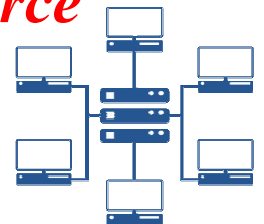
HYPER TEXT TRANSFER PROTOCOL-80



What is HTTP Protocol?



- ✓ **Hypertext Transfer Protocol (HTTP)** is a method for encoding and transporting information between a client (such as a web browser) and a web server.
- ✓ HTTP is the primary protocol for transmission of information across the LAN/WAN.
- ✓ Port no is **80.(TCP)**
- ✓ HTTP resources such as web servers are identified across the Internet using unique identifiers known as ***Uniform Resource Locators (URLs)***.



Available Web server Product?

Product	Vendor	Percent
Apache	Apache	44.3%
Nginx	NGINX,Inc	41.0%
IIS	Microsoft	8.9%
LiteSpeed Web Server	LiteSpeed Technologies	3.9%
GWS	Google	0.9%

Requirements for Webserver?



1. System with Windows Operating System (Client OS or Server OS)

2. HTTP Server/Web Server

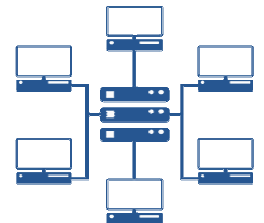
IIS-Internet Information Service.

Internet Information Services is an **extensible web server software** created by Microsoft for use with the Windows NT family.

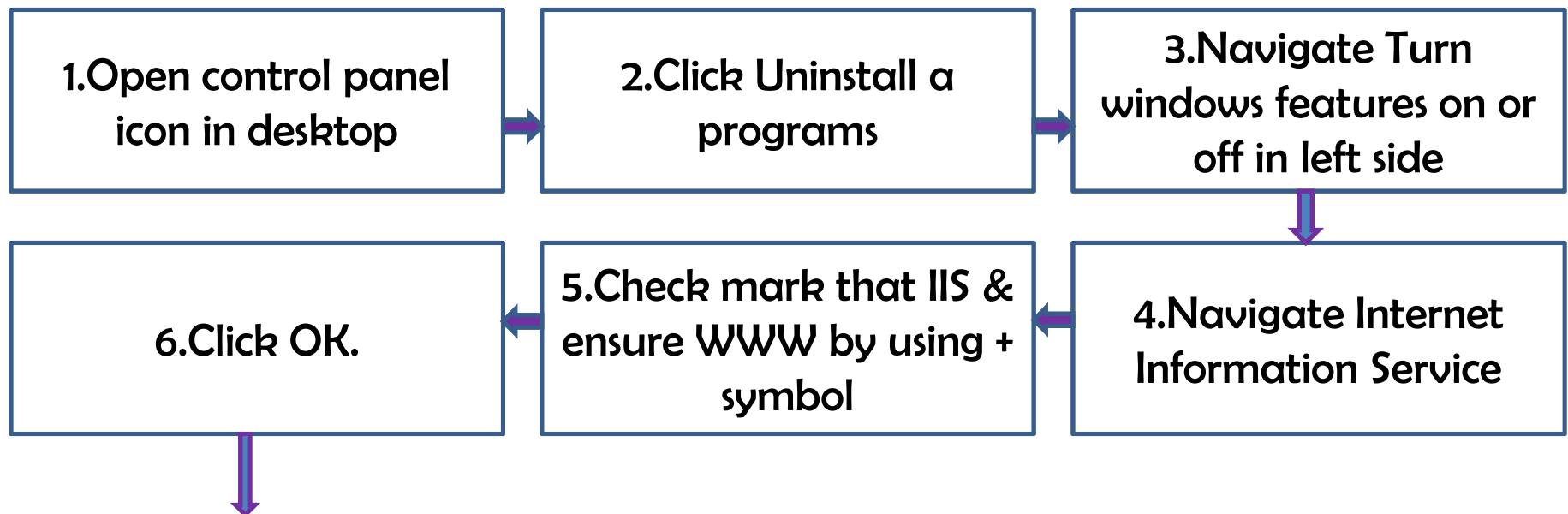
Internet Information Services (IIS) for Windows Server is a **flexible, secure and manageable Web server** for hosting anything on the Web.

IIS supports HTTP, HTTP/2, HTTPS, FTP, FTPS, SMTP and NNTP.

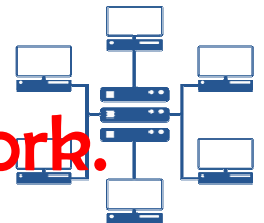
3. Webpage



How to Install HTTP Protocol?



Now this machine will be able to host website and remotely from another computer over the network.



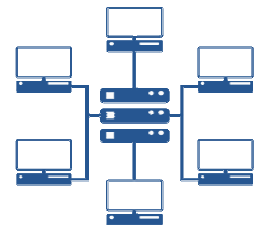
How to configure Web Server in Windows Server?

✓ After Installation of IIS, one folder created.

Path- C:\inetpub

wwwroot-by default webpage

✓ If we can control this IIS by using Internet Information Service manager tool.

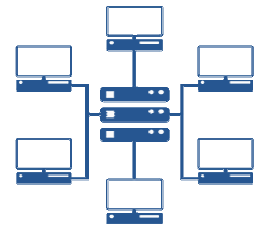


How can you Access HTTP Protocol?

Pre-requisites: **URL-Uniform Resource Locator(//)**
Software- **Web Browsers**

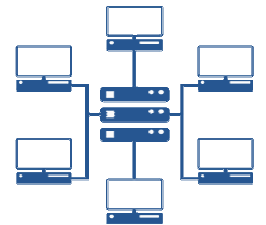


Private IP/ Public IP-
Type this IP Address in Address bar



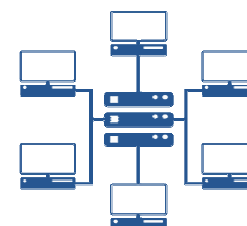


Are You Connected with Internet Connection?



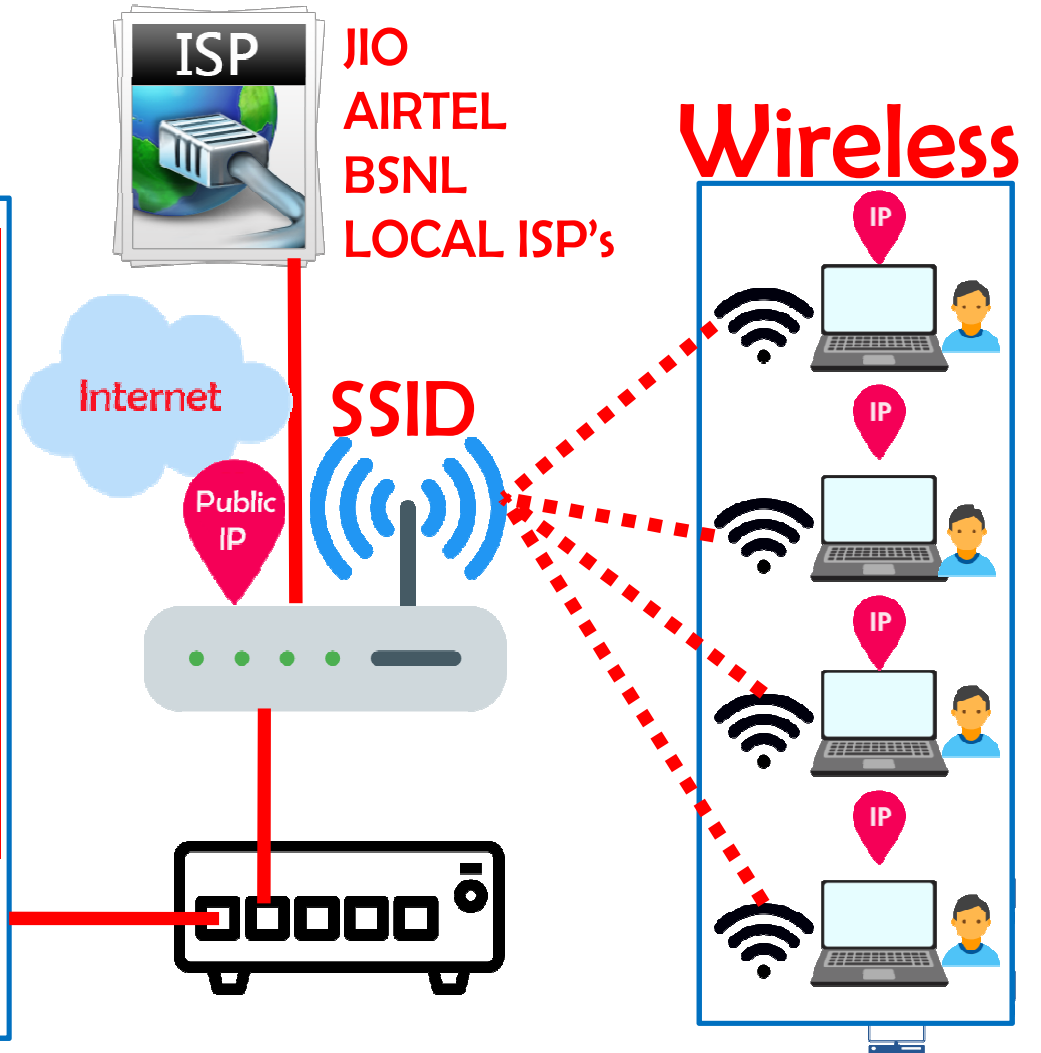
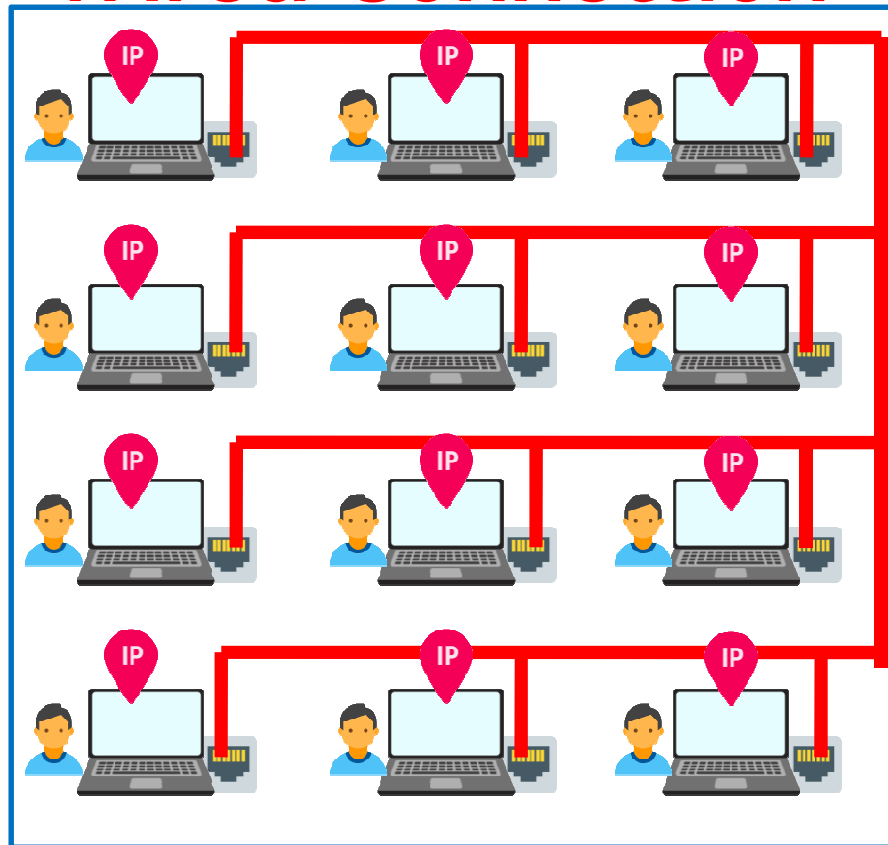


How to connect Internet connection?



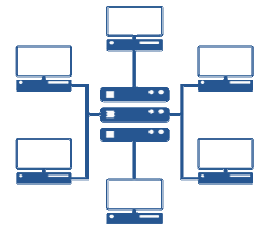
Our Scenario

Wired Connection



Demo

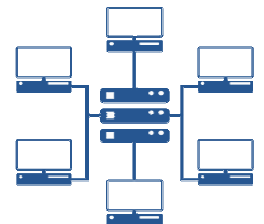
HOW TO CONNECT INTERNET?



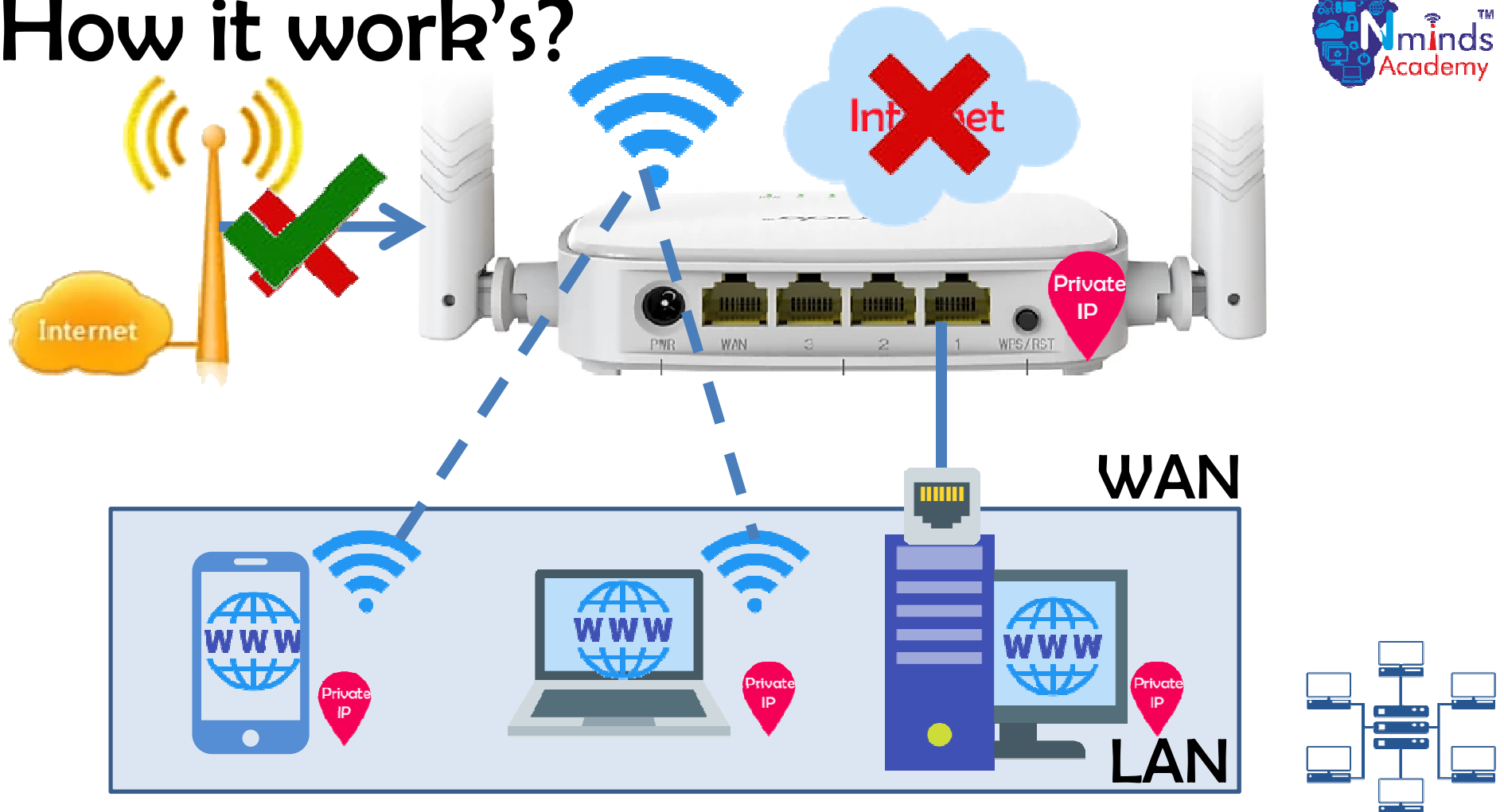
What is WISP?



- ✓ WISP- **Wireless Internet Service Provider.**
- ✓ A wireless Internet service provider (WISP) is an Internet service provider (**ISP**) that allows subscribers to connect to a server at designated **hot spots** (access points) using a **wireless connection such as Wi-Fi.**

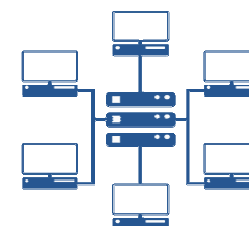


How it work's?

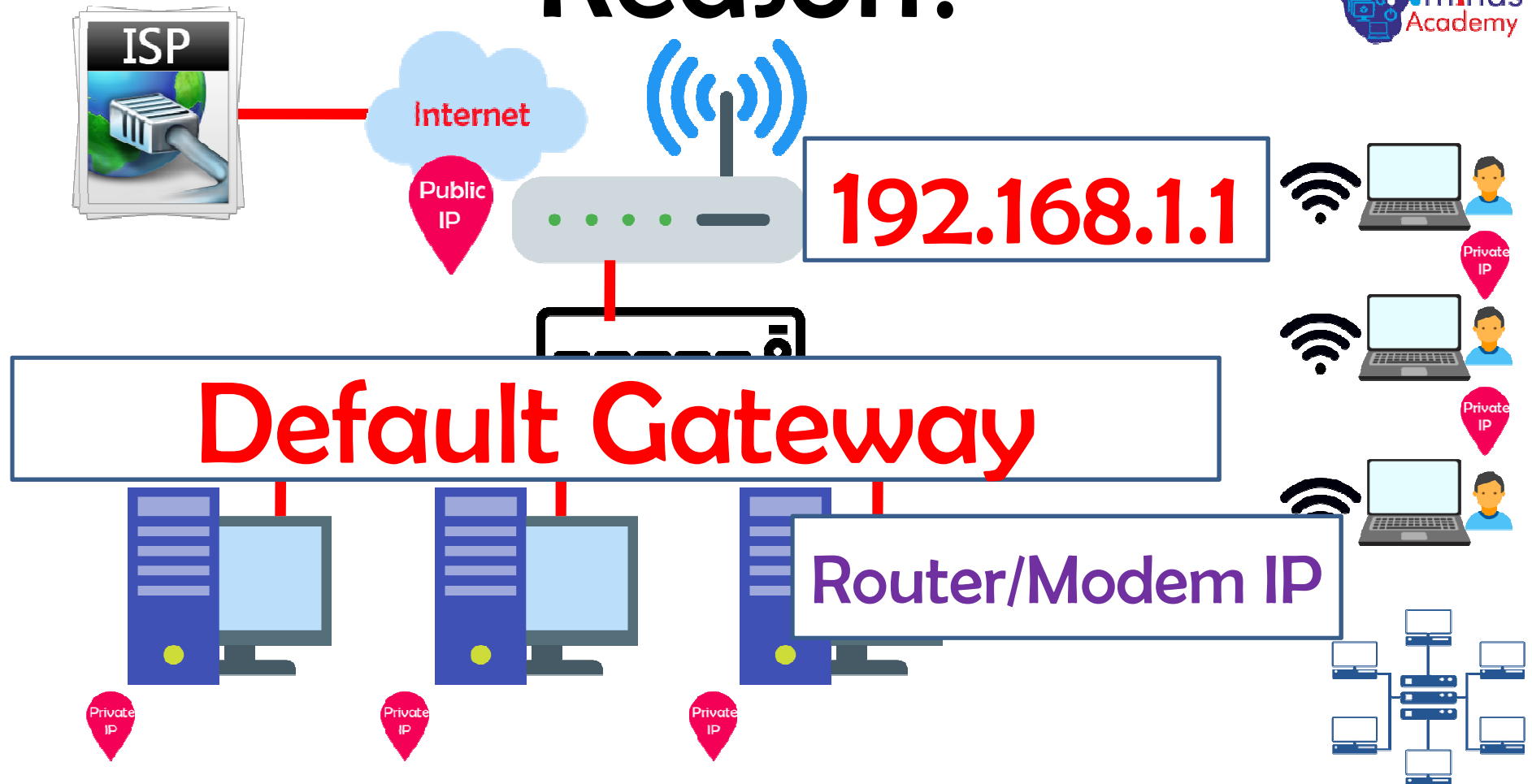




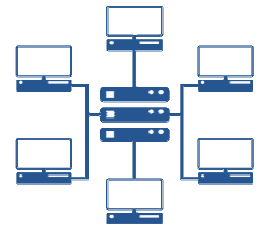
Is internet connect to your machine?



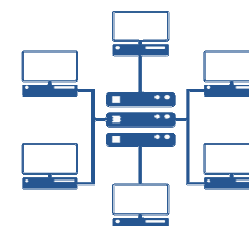
Reason?



Proof-Default Gateway



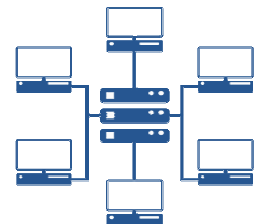
DHCP Configuration



What is DHCP?



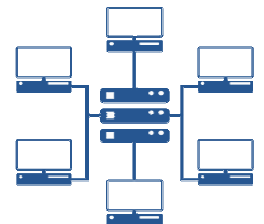
- ✓ DHCP-Dynamic Host Configuration Protocol
- ✓ DHCP is a client/server protocol that automatically provides an Internet Protocol (IP) host with its IP address and other related configuration information such as the subnet mask and default gateway.
- ✓ DHCP Server port-67 client port-68(UDP)



DHCP configuration in Router?

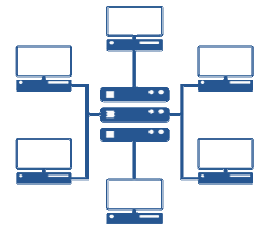


- ✓ DHCP server is ON in all Router/Modem by default.
- ✓ Lease time-How long the IP address present in that machine.
- ✓ When DHCP is failure/unavailable by default IP address is assigned in every machine.
- ✓ That IP address is called APIPA(Automatic Private IP Address)-169.254.XX.XX

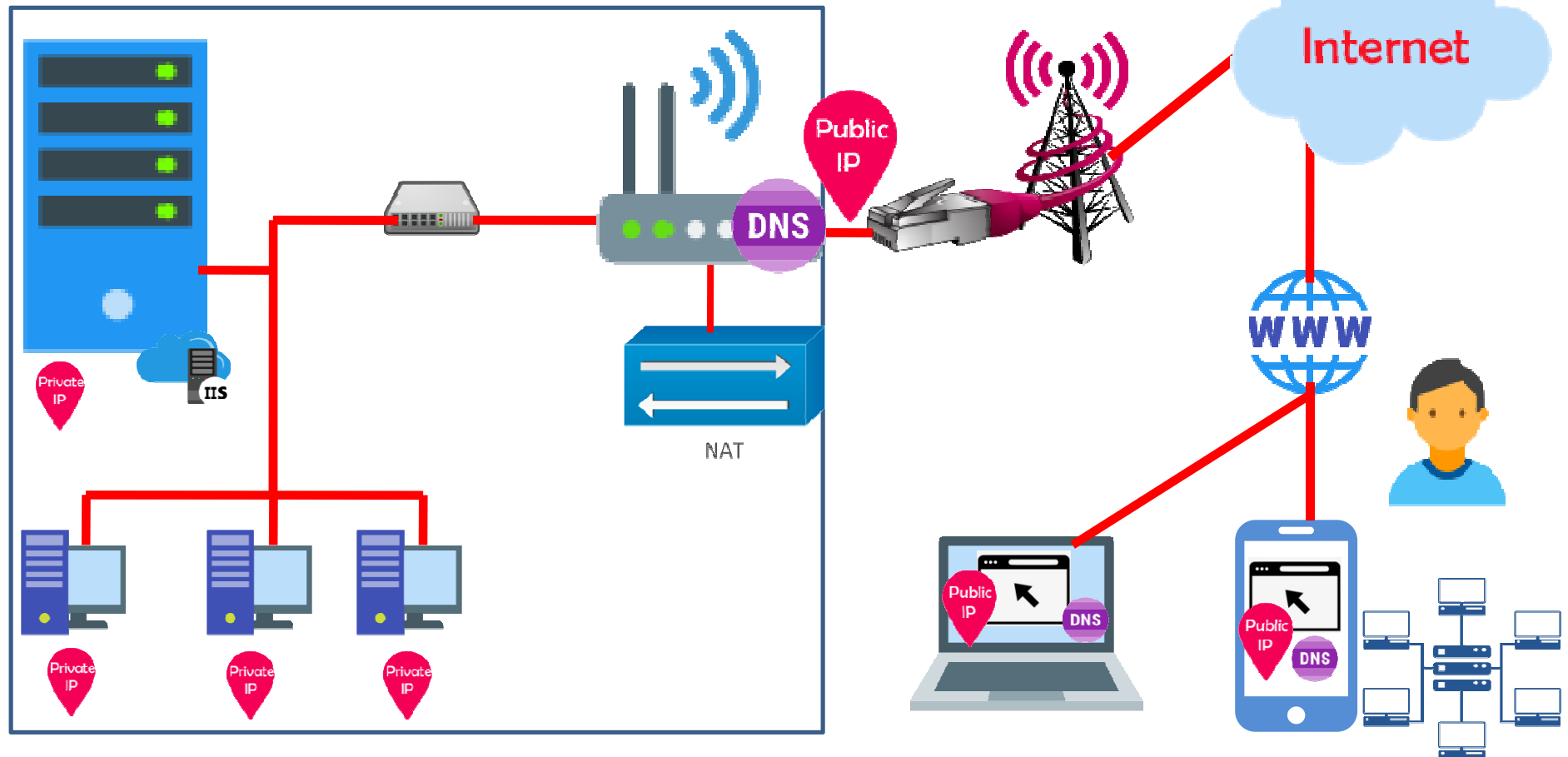


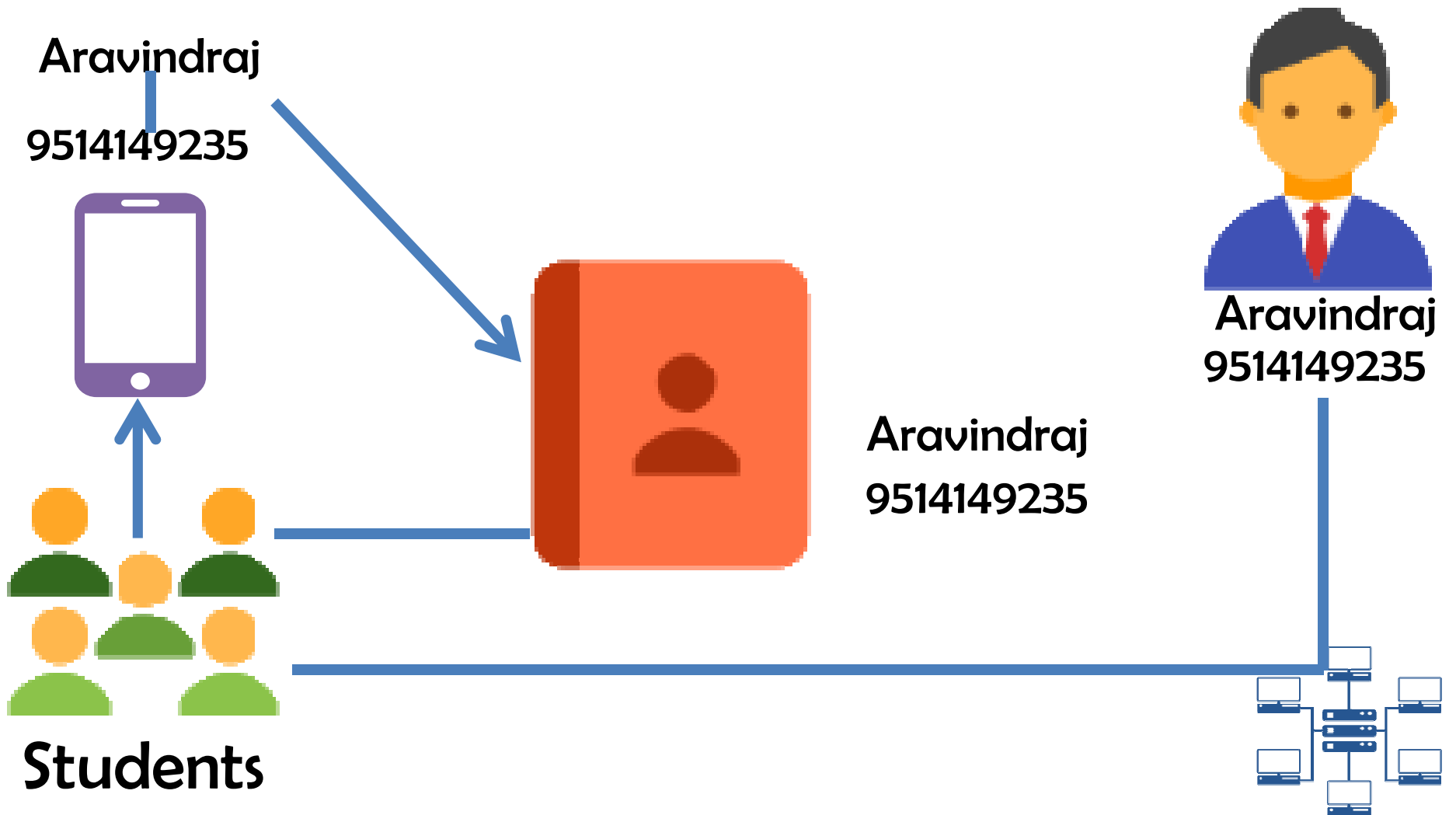
Demo

HOW WEBSITE RUN IN WAN?

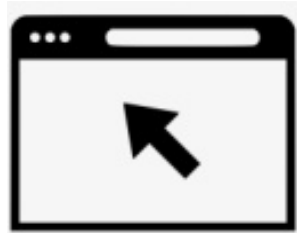


HOW WORKS ON-PREMISE WEB SERVER?



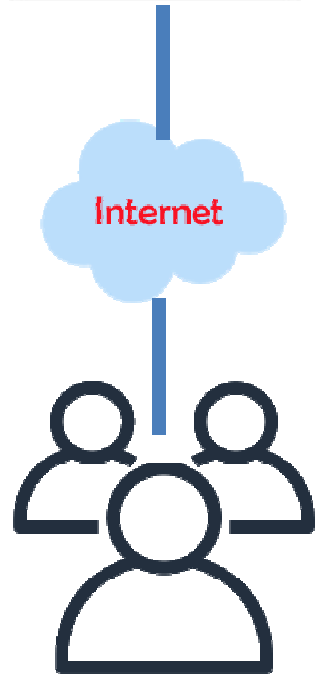


Need of DNS in Internet?

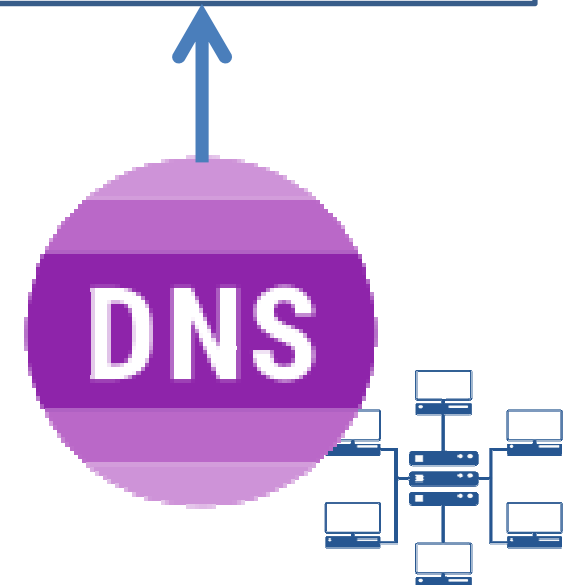


142.250.67.78	google.com
157.240.23.35	facebook.com
72.163.4.185	cisco.com

**DNS resolves Domain
name to IP address**

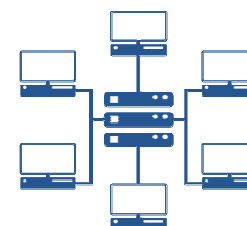


google.com	142.250.67.78
facebook.com	157.240.23.35
cisco.com	72.163.4.185



DNS

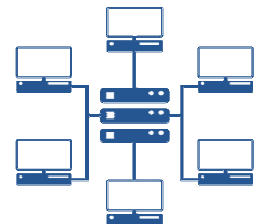
DOMAIN NAME SERVER-53



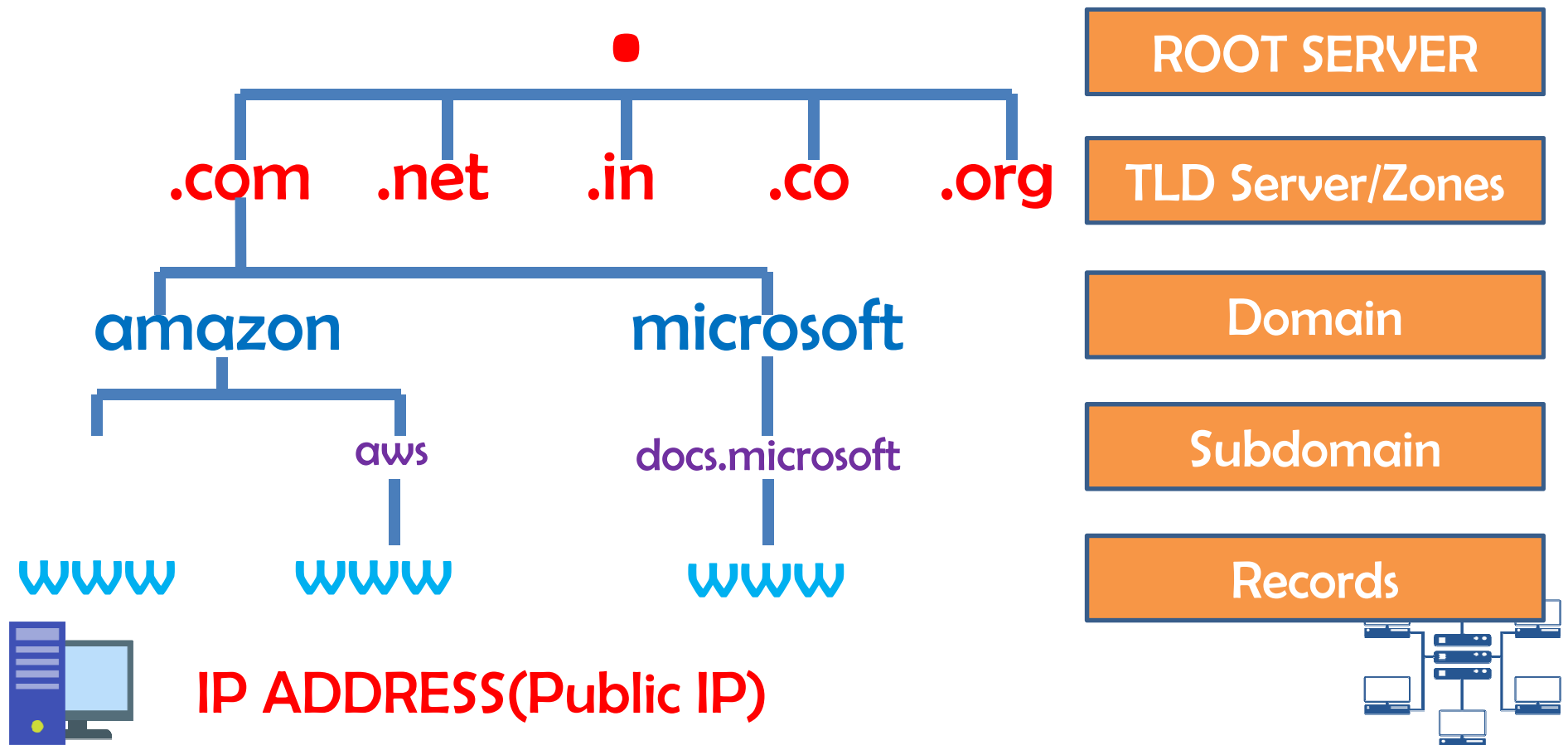
What is DNS Protocol?



- ✓ Domain Name Server (**DNS**) is a standard protocol that helps Internet users discover websites using human readable addresses.
- ✓ Web browsers interact through Internet Protocol (IP) addresses.
- ✓ DNS translates domain names to IP addresses so browsers can load Internet resources.
- ✓ The Domain Name System (**DNS**) is the phonebook of the Internet. Humans access information online through domain names.
- ✓ DNS is UDP Protocol and port no is 53.



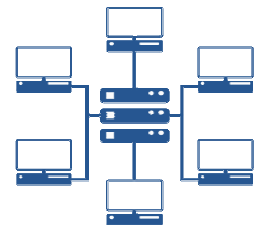
Hierarchy Structure Of DNS



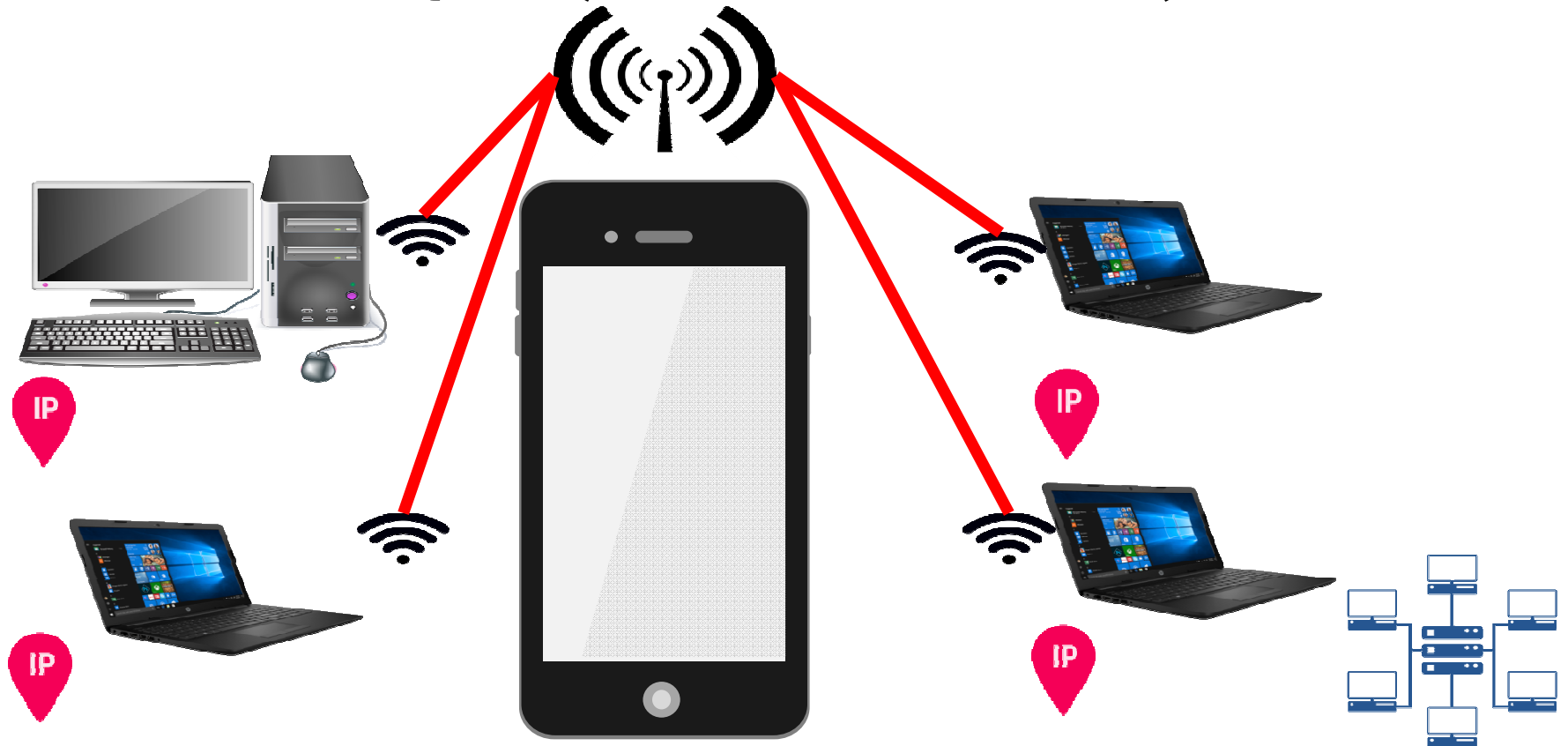


Capstone Project

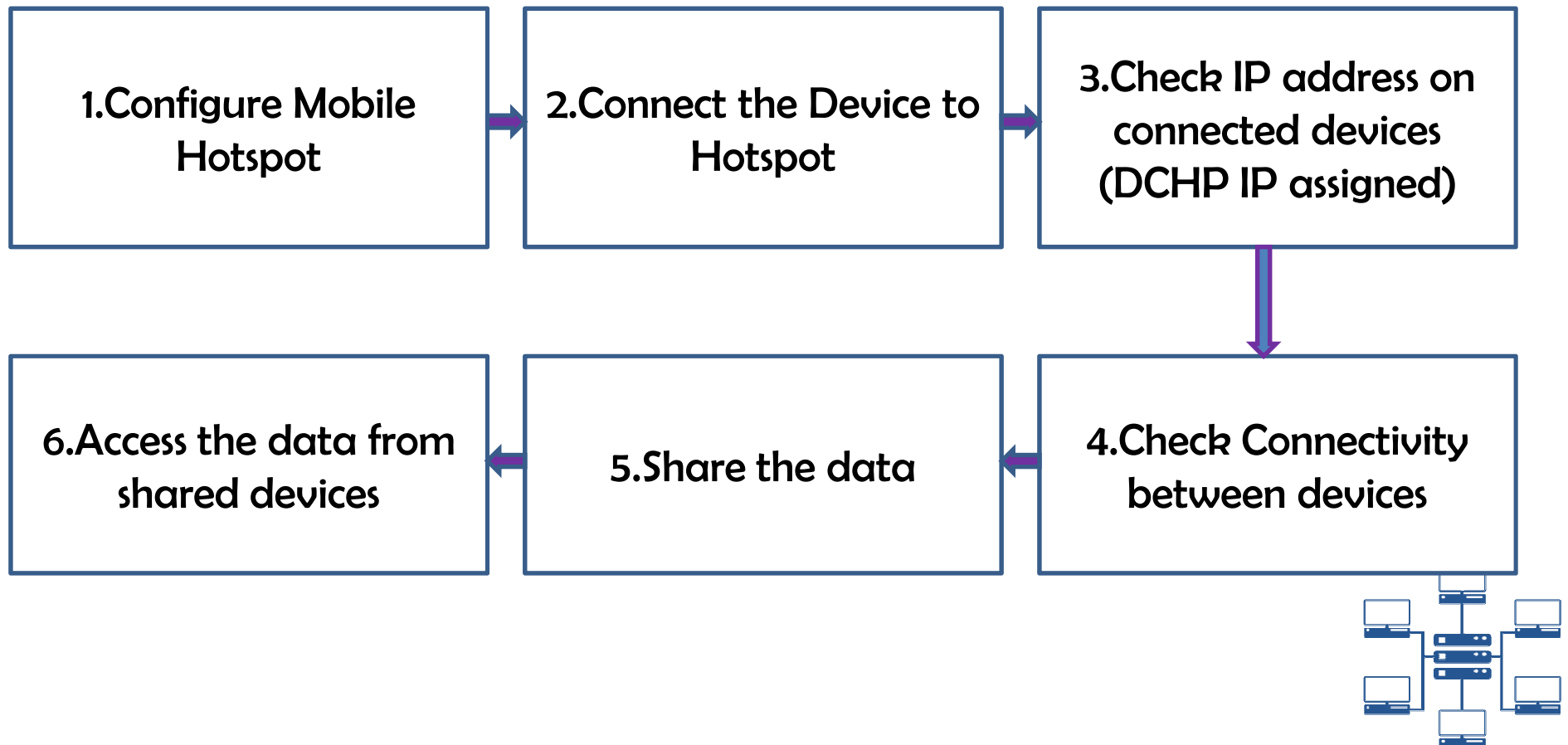
TRY IT YOUR SELF



Multipoint Connection by using Mobile Hotspot(Wireless Media)



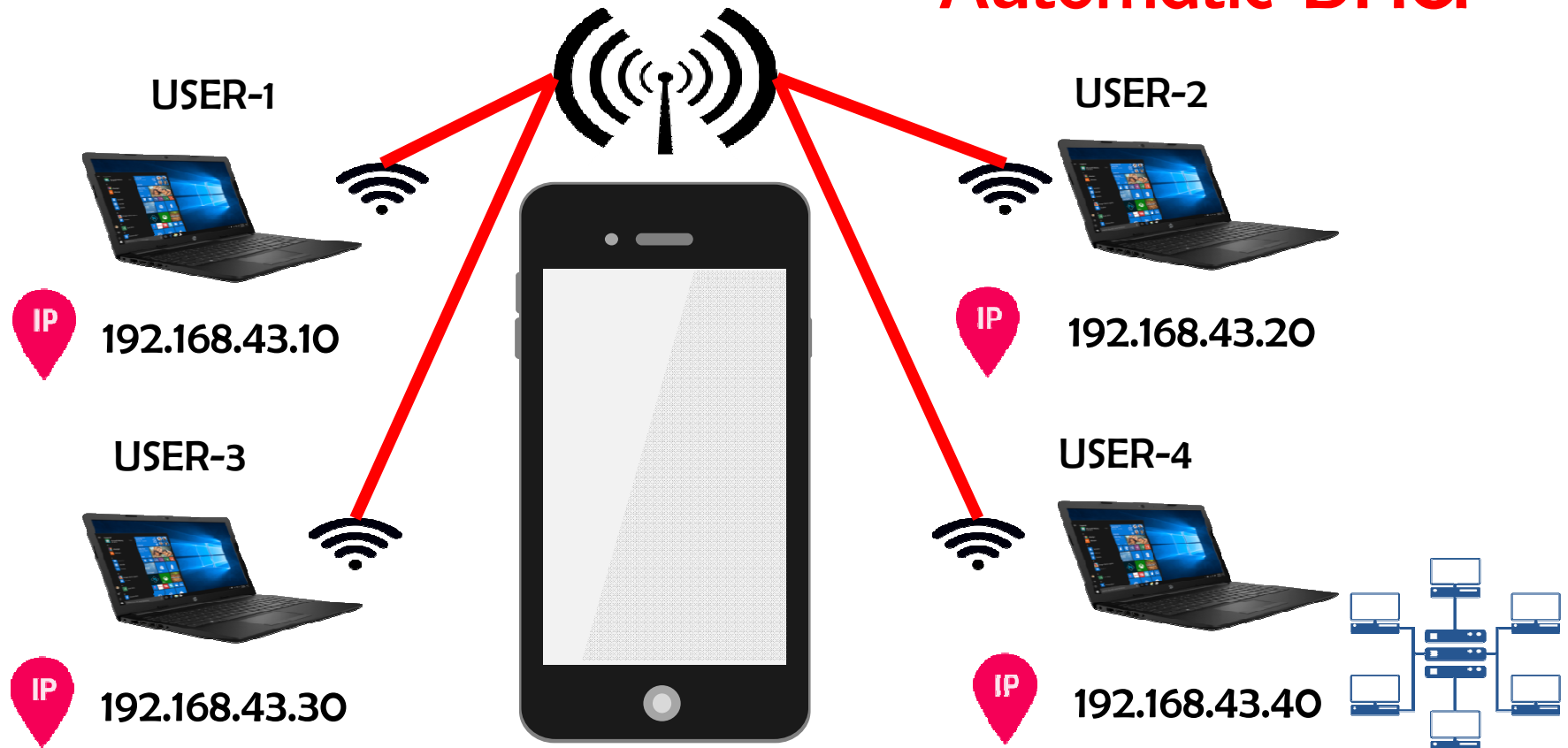
Steps to Connect Device via Hotspot?

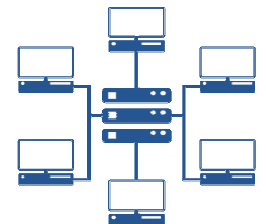


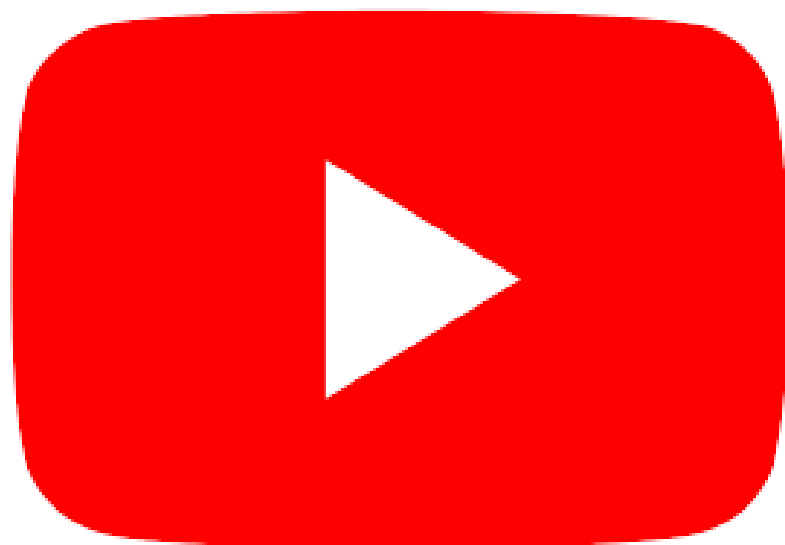
How Mobile Hotspot Works?



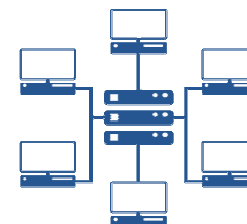
NMINDS-Hotspot Automatic-DHCP



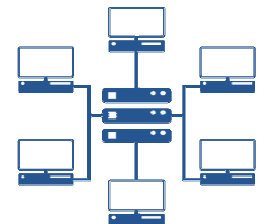
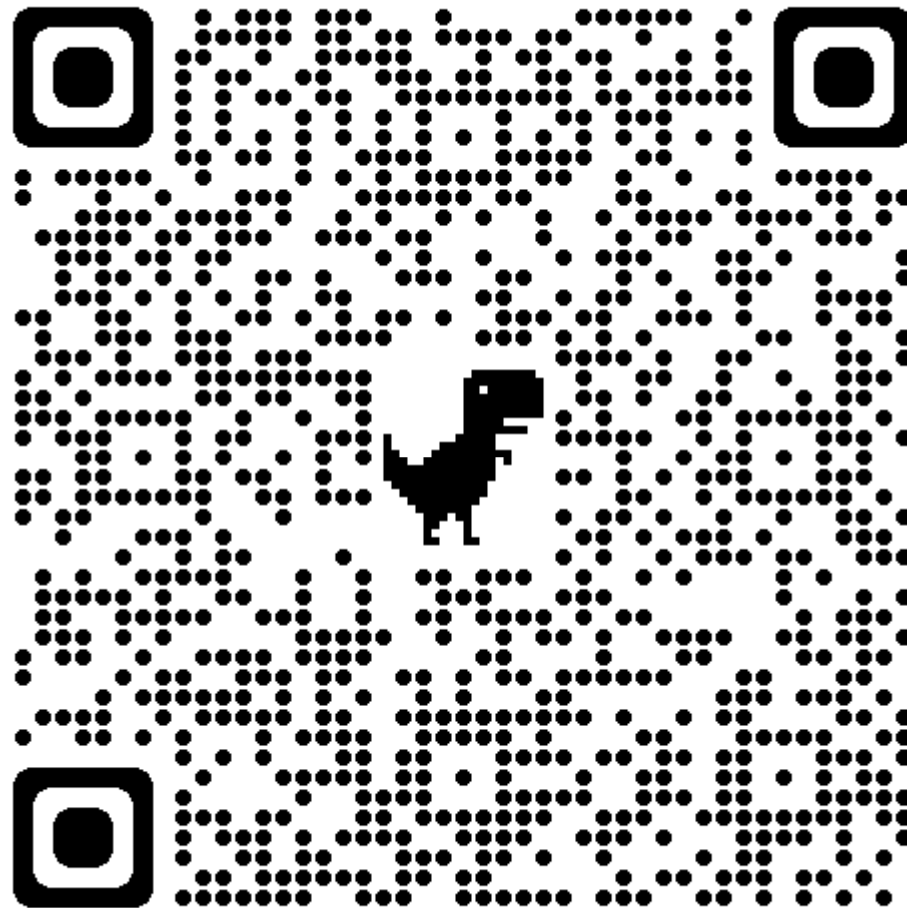




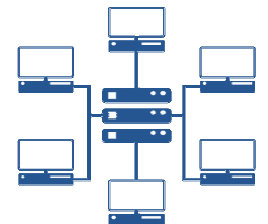
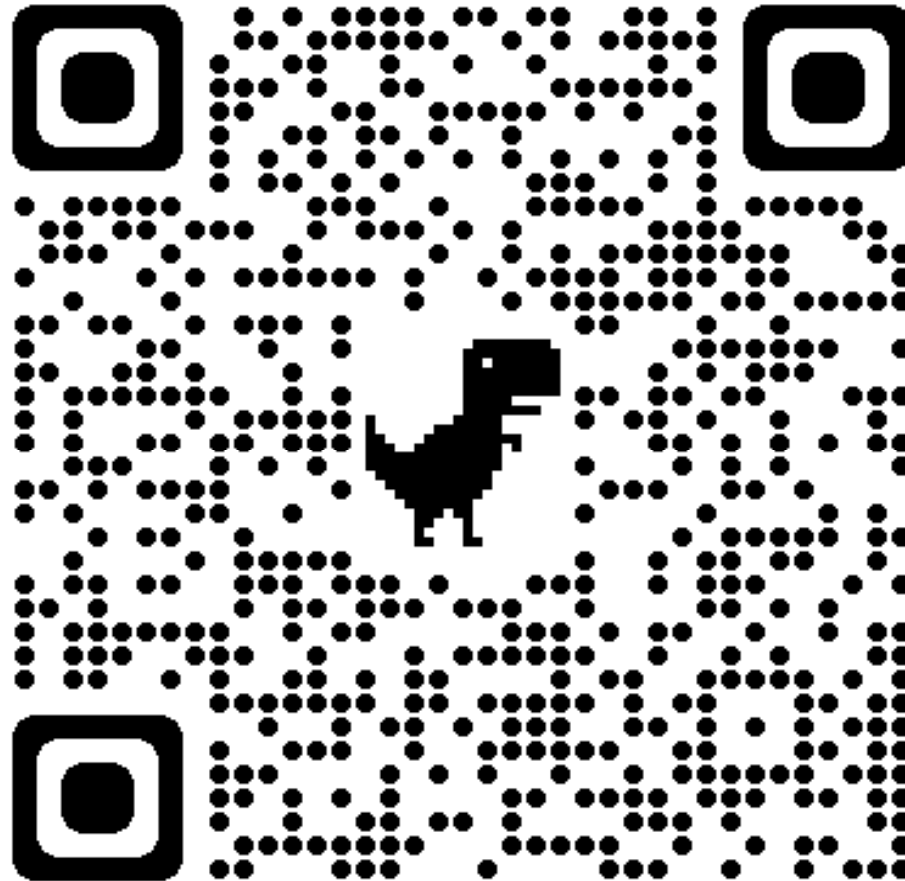
NmindsAcademy



Quiz



Feedback



Thank you!



The Expert in Everything Was Once a Beginner...



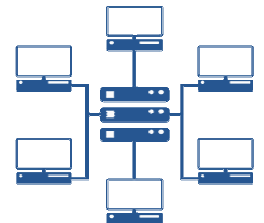
ARAVINDRAJ.G  95141 49235
Proprietor

- ▶ Training
- ▶ Workshop
- ▶ Seminar



19, Indira Nagar, Narayanavalasu, Erode - 638 011.

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Section End

We Say Thank you for Your Response